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Month, Year

To my family.

Acknowledgments

I would like to thank my supervisor, **Prof. Supervisor Name**, for the guidance and support throughout this work.

Resumo

Este trabalho aborda o problema de …

Palavras-chave: palavra-chave 1, palavra-chave 2, palavra-chave 3.

Abstract

This thesis addresses the problem of …

Keywords: keyword 1, keyword 2, keyword 3.

Contents

List of Acronyms

Acronym	Definition
ACDC	Automated Cardiac Diagnosis Challenge
AHA	American Heart Association
ASSD	Average Symmetric Surface Distance
DSC	Dice Similarity Coefficient
DCM	Dilated Cardiomyopathy
ED	End-Diastole
EDT	Euclidean Distance Transform
ES	End-Systole
GNN	Graph Neural Network
HCM	Hypertrophic Cardiomyopathy
HD95	95th-percentile Hausdorff Distance
ICC	Intraclass Correlation Coefficient
INR	Implicit Neural Representation
IoU	Intersection over Union
LV	Left Ventricle
LoA	Limits of Agreement
MAE	Mean Absolute Error
MRI	Magnetic Resonance Imaging
NOR	Normal cohort
PCA	Principal Component Analysis
PDE	Partial Differential Equation
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RMSE	Root Mean Squared Error
RV	Right Ventricle
SAX	Short-Axis
SDF	Signed Distance Function
SSM	Statistical Shape Model

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CHAPTER 1

Introduction

This chapter introduces the clinical and technical motivation for reconstructing three-dimensional left-ventricular geometry from sparse cardiac magnetic resonance imaging (MRI). It defines the reconstruction problem, presents the research questions and objectives, and summarises the main contributions of this thesis.

1.1. Motivation

Understanding cardiac structure is fundamentally a geometric problem. The left ventricle (LV) is a continuously varying three-dimensional structure whose shape, size, and wall thickness change throughout the cardiac cycle. These properties provide important information about cardiac function and structural remodelling, with changes in ventricular geometry and myocardial thickness being associated with several forms of cardiac disease (de Marvao et al., 2014; Grossman et al., 1974; Medrano-Gracia et al., 2014).

Cardiac magnetic resonance imaging (MRI) provides detailed observations of this anatomy, but these observations are acquired as a set of two-dimensional slices rather than as a complete three-dimensional representation. In a short-axis (SAX) acquisition, the LV is sampled from the base towards the apex, allowing the endocardial and epicardial boundaries to be delineated on individual slices. Within each slice, the spatial resolution can be relatively high, while the distance between adjacent slices is substantially larger. Consequently, a structure that is anatomically continuous is represented by a limited number of discrete observations.

This distinction becomes important when measurements are extended beyond the acquired slices. Myocardial wall thickness, for example, is not simply a property of isolated contours. It varies across the ventricle and can reflect changes in cardiac structure, including myocardial hypertrophy (Grossman et al., 1974). Estimating such quantities from sparse observations therefore depends not only on the accuracy of the segmentation, but also on how well the underlying three-dimensional geometry is represented.

A continuous reconstruction can provide a more coherent description of the LV and enable measurements in regions that are not directly observed. Such representations are particularly relevant in large imaging studies, where consistent anatomical descriptions are needed for automated analysis and comparison across subjects (Bai et al., 2015, 2018). They can also support regional analysis of myocardial structure, including measurements organised according to the American Heart Association 17-segment model (Cerqueira et al., 2002).

The central challenge addressed in this thesis is therefore not the extraction of LV contours alone, but the inference of the three-dimensional geometry that lies between sparse observations. A useful reconstruction must recover continuous endocardial and epicardial surfaces while preserving patient-specific shape and maintaining a anatomically plausible myocardial wall. Simple interpolation or surface lofting can provide continuity, but does not necessarily recover the geometry that best represents the underlying anatomy.

This thesis investigates whether a phase-conditioned implicit neural representation can learn this reconstruction problem from synthetic LV geometries and meshes derived from real cardiac MRI segmentations. The proposed approach represents the LV surfaces as continuous functions in three-dimensional space, allowing geometry to be queried at arbitrary locations rather than only at the acquired slices. The reconstructed surfaces are subsequently used to study myocardial wall thickness and to compare the resulting measurements with those obtained from RBF-derived clinical geometry. The aim is not to replace clinical interpretation, but to investigate whether continuous geometric representations can provide a more consistent basis for quantitative analysis of sparse cardiac MRI.

1.2. Problem Statement

SAX cardiac MRI provides detailed cross-sectional information about the LV, but does not directly provide a complete three-dimensional representation. The slices are separated along the ventricular long axis, leaving portions of the geometry unobserved between adjacent acquisitions. As a result, the spatial resolution within each slice is typically much finer than the resolution between slices. Reconstructing a continuous surface from these sparse observations is therefore an ill-posed geometric problem.

A simple connection of the segmented contours may produce a continuous surface, but continuity alone does not guarantee a plausible reconstruction. Such approaches can introduce irregularities, gaps, or local distortions, especially when the distance between slices is large. The challenge is particularly important when both the endocardial and epicardial surfaces are considered, because they must be reconstructed as a consistent pair. The endocardium defines the inner boundary of the LV cavity, while the epicardium defines the outer boundary of the myocardium. Their relative position must remain physiologically plausible so that the reconstructed wall retains positive and spatially coherent thickness.

The problem is further constrained by the characteristics of publicly available cardiac MRI datasets. These datasets commonly provide images and segmentation masks for individual slices rather than complete three-dimensional LV meshes. Meshes are nevertheless useful for directly training and evaluating surface-reconstruction models because they provide an explicit representation of the continuous geometry. A method is therefore required to bridge the gap between sparse slice-based observations and continuous three-dimensional surfaces.

This thesis addresses this problem using a phase-conditioned implicit neural representation (INR) trained on synthetic LV shapes and meshes derived from real cardiac MRI segmentations. The model receives sparse contours as input and represents the endocardial and epicardial surfaces as continuous three-dimensional functions. Conditioning the representation on cardiac phase allows the model to distinguish between end-diastolic and end-systolic geometry. In addition, the reconstruction is constrained so that the outer surface remains outside the inner surface, preserving a positive myocardial wall thickness.

The reconstructed geometry is evaluated at both the surface and measurement levels. Geometric agreement is assessed against reference surfaces, while myocardial wall thickness is measured on the reconstructed geometry and compared with measurements obtained from RBF-derived clinical geometry. This makes it possible to evaluate not only whether the proposed model produces visually continuous surfaces, but also whether the resulting geometry preserves quantities relevant to cardiac analysis.

1.3. Research Questions

This thesis addresses the following research questions:

RQ1. To what extent can the proposed model reconstruct continuous, watertight endocardial and epicardial LV surfaces from sparse SAX contours, how closely does it follow the observed contours, and how far does it agree with conventional implicit and shape-model reconstructions of the same contours?

RQ2. Can the phase-conditioned model represent end-diastolic and end-systolic LV geometry while maintaining a positive myocardial-wall separation between the endocardial and epicardial surfaces?

RQ3. To what extent do local and regional wall-thickness measurements on the reconstructed geometry agree with those obtained from RBF-derived clinical geometry, and how does that agreement depend on the measurement method?

CHAPTER 2

Literature Review

This chapter describes the review methodology and examines prior work on reconstructing paired LV surfaces from sparse cardiac MRI observations and measuring the myocardial wall represented by those surfaces.

2.1. Review Methodology

The review was designed to compare methods relevant to sparse cardiac MRI, three-dimensional shape reconstruction, geometric learning, and myocardial wall-thickness measurement, rather than to provide an exhaustive systematic review.

Scopus, PubMed, and IEEE Xplore were searched, with Google Scholar used for backward and forward citation chaining. The search included foundational work published from 2000 onwards and selected studies available up to 2026. Although recent work was prioritised, earlier publications were retained where they establish concepts used by later methods. The resulting review is selective and does not claim exhaustive coverage of the literature published through 2026.

Search terms combined anatomy and imaging concepts (“cardiac MRI”, “left ventricle”, “short-axis”, “SAX”, “myocardium”) with reconstruction terms (“3D reconstruction”, “mesh reconstruction”, “shape model”, “implicit surface”, “signed distance function”), learning terms (“deep learning”, “graph neural network”, “point cloud”, “neural representation”), and measurement terms (“wall thickness”, “myocardial thickness”, “Laplace equation”, “transmural”).

The refined query applied consistently across databases was:
(“cardiac MRI” OR “left ventricle” OR “short-axis”) AND (“3D reconstruction” OR “statistical shape model” OR “signed distance function” OR “graph neural network” OR “implicit representation”) AND (“wall thickness” OR “mesh” OR “deep learning”)

Studies were included when they addressed LV three-dimensional reconstruction or shape modelling, proposed or evaluated myocardial wall-thickness methods, developed geometric learning methods applicable to anatomical meshes, or introduced benchmark cardiac MRI data relevant to LV analysis. Studies focused only on segmentation, on non-cardiac anatomy without transferable methodology, or lacking peer review or sufficient methodological detail were excluded. Publications before 2000 were considered only when they provided foundational methods directly relevant to the reviewed approaches.

The search returned 142 records. After duplicate removal and screening, 28 studies met the selection criteria; backward citation searching added 7, giving a final corpus of 35 publications. Figure 2.1 summarises this process.

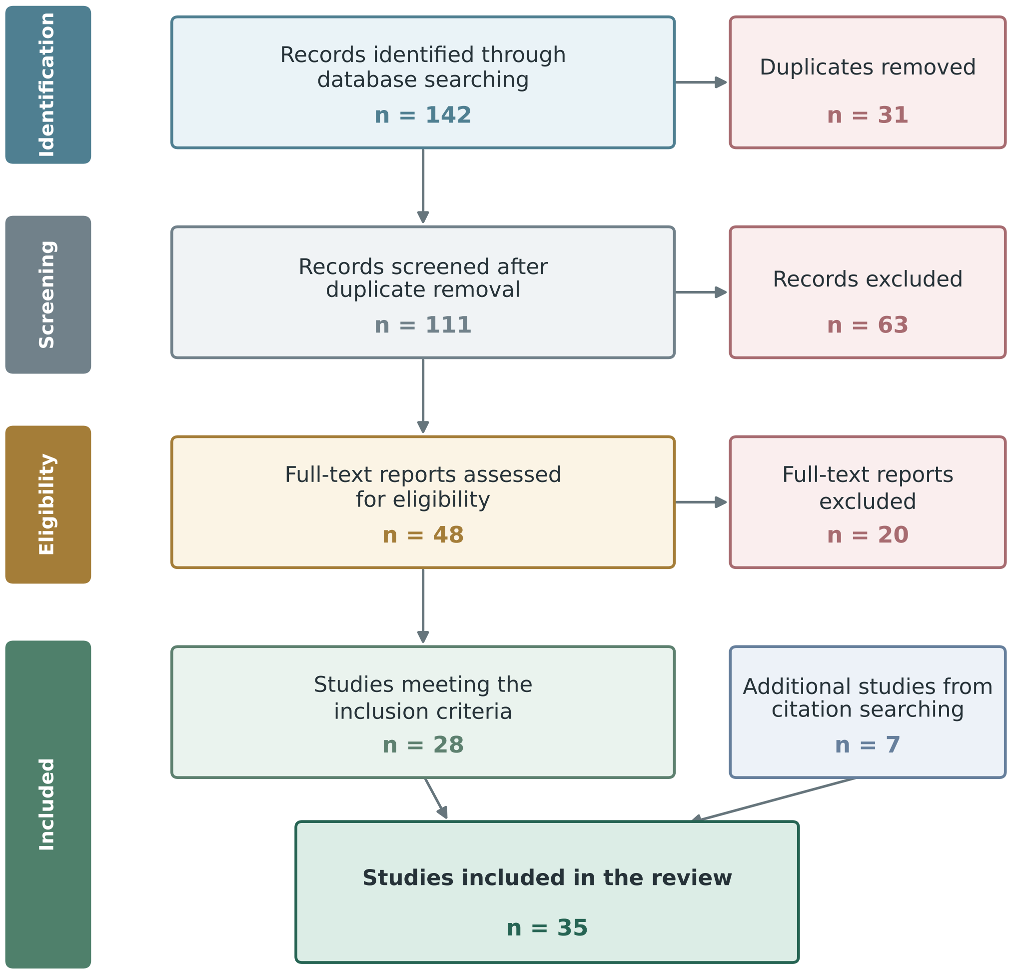

Figure 2.1. PRISMA flow diagram summarising the selection of the 35 publications included in the review.

The selected studies cover cardiac MRI data and segmentation, statistical shape models, learned 3D reconstruction, and wall-thickness measurement. The next section relates these areas to the specific problem of recovering paired LV surfaces from sparse contours.

2.2. Technical Background

The central distinction in this review is between anatomy observed on acquired MRI slices and geometry that must be inferred between them. This distinction connects the constraints of paired LV surface reconstruction to the available representations of three-dimensional anatomy.

Cine cardiac MRI captures the LV at several points in the cardiac cycle. Of these, end-diastole (ED), when the cavity is largest, and end-systole (ES), when it is smallest, provide a useful description of ventricular contraction and are the phases labelled in benchmarks such as ACDC (Bernard et al., 2018).

In a SAX acquisition, each plane samples a cross-section perpendicular to the ventricular long axis. The anatomy is resolved in detail within a slice, but the larger separation between consecutive slices produces an anisotropic image stack (Petitjean & Dacher, 2011). Segmentation can therefore recover the myocardium where it is observed without determining how its boundaries curve, taper, or change in the unobserved intervals.

Each slice contributes an endocardial contour around the cavity and an epicardial contour around the outer myocardium. In three dimensions, these must form nested surfaces with a positive separation. A reconstruction that treats the surfaces independently may produce individually plausible boundaries while still creating an intersection, reversal, or local collapse in the wall between them, a difficulty also addressed by recent joint implicit reconstructions of cardiac structures (Brás et al., 2026).

Recovering the continuous geometry is consequently underdetermined: many smooth surfaces can pass through the same contour rings. Smoothing stabilises the reconstruction but can remove genuine regional variation; a strong population prior can bridge missing regions but may pull abnormal anatomy towards common shape; and close contour fitting can preserve segmentation noise. A useful method must balance these competing constraints rather than optimise any one of them in isolation (Heimann & Meinzer, 2009; Petitjean & Dacher, 2011).

The same reasoning determines how reconstruction should be evaluated. The surfaces should agree with observed anatomy, remain closed where volume is needed, preserve subject-specific shape, and form a valid wall at both phases. No single overlap or surface-distance measure assesses all of these properties, so geometric distance, volumetric agreement, topology, and the stability of wall-derived measurements must be considered together (Brás et al., 2026; Heimann & Meinzer, 2009).

The representation chosen for three-dimensional anatomy determines both what can be measured and how missing geometry is inferred. Voxel volumes are well suited to image-based segmentation and overlap analysis, but retain the spatial limits of the sampling grid. Surface meshes support direct geometric and regional measurements, although remeshing, smoothing, and repair can alter the anatomy being evaluated. Neither representation alone determines the geometry between sparsely acquired slices.

Statistical shape models address this missing information by constraining meshes with population-level modes of variation (Cootes et al., 1995; Heimann & Meinzer, 2009). They can therefore provide a geometric prior or synthetic training examples, but remain limited by the topology and anatomical variation represented in the source atlas.

Implicit methods offer a different compromise. DeepSDF and Occupancy Networks show that geometry can be represented continuously and converted to a mesh at a chosen resolution (Lorensen & Cline, 1987; Mescheder et al., 2019; Park et al., 2019). Work on Eikonal regularisation and Fourier features further addresses distance-field consistency and the recovery of local spatial variation (Gropp et al., 2020; Tancik et al., 2020). By improving field consistency and spatial detail, these developments make implicit representations relevant to sparse-contour reconstruction. The extracted surface nevertheless remains dependent on the learned field and extraction procedure.

Paired LV surfaces introduce an additional constraint: two independently predicted fields do not guarantee a valid ordering. Defining one surface in relation to the other through a positive separation instead represents a reference boundary and a constrained wall. Joint implicit reconstruction of cardiac structures provides one way of modelling related boundaries together, although surface ordering and wall validity must still be assessed explicitly (Brás et al., 2026).

2.3. Related Work

This section compares how the reviewed methods address the evidence available on acquired slices, the geometry missing between them, and the measurements derived from the reconstructed wall.

2.3.1. Clinical observations and conventional geometric priors

Short-axis (SAX) cine MRI is the standard protocol for left-ventricle (LV) evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) (Cerqueira et al., 2002; Petitjean & Dacher, 2011). The resulting anisotropy creates the central reconstruction problem: the observed contours are detailed within each slice, but the geometry between slices, especially near the apex and base, must be inferred. The AHA 17-segment model provides a common anatomical frame for reporting regional measurements (Cerqueira et al., 2002).

Modern segmentation networks can identify endocardial and epicardial boundaries reliably on the acquired slices. U-Net established the encoder–decoder pattern used by many medical segmentation systems, while nnU-Net showed that much of their configuration can be adapted automatically to a dataset (Isensee et al., 2021; Ronneberger et al., 2015). Although both methods improve the observed contours, they still optimise agreement on sampled image planes. High slice-wise overlap can therefore coexist with uncertain through-plane geometry. Moreover, because the contours become constraints on the reconstructed surface, segmentation errors can propagate into regions where no image evidence is available rather than remaining local to one slice.

ACDC provides expert ED and ES labels across five diagnostic groups, including normal and hypertrophic ventricles (Bernard et al., 2018). Its phase and diagnostic variation make it useful for testing whether a reconstruction retains contraction-related and abnormal shape differences. The M&Ms-2 challenge focuses on right-ventricular segmentation and extends evaluation across centres, scanner vendors, and diseases (Martín-Isla et al., 2023). For reconstruction, its multi-source design shows that performance on one acquisition source does not guarantee transfer to another. A plausible surface on one cohort is insufficient evidence of generalisation because acquisition variation affects the contours from which the entire surface is inferred.

At population scale, the UK Biobank provides a large and standardised cardiac MRI resource (Sudlow et al., 2015). Automated segmentation of this cohort has enabled large anatomical studies and statistical atlases (Bai et al., 2018). Together, slice-wise clinical observations and population-level atlases provide complementary forms of anatomical evidence.

The two forms are not interchangeable. Clinical labels retain subject-specific and pathological variation, but their through-plane geometry is only indirectly observed. A population atlas supplies complete and consistent surfaces, but it reflects the cohort and processing choices from which it was built. Combining them can reduce the weakness of either source alone, provided that the atlas is used as a prior rather than treated as patient-specific ground truth.

Statistical Shape Models (SSMs) encode anatomical variability by aligning a training population to point-to-point correspondence and applying PCA to extract the dominant modes of variation (Cootes et al., 1995; Heimann & Meinzer, 2009). They provide a compact description of plausible shape, but their quality depends on anatomical correspondence, alignment, and the diversity of the population used to construct them.

Cardiac SSMs have progressed from multi-object ventricular models to large population atlases. Dense correspondence allows coupled structures to be represented consistently across subjects (Frangi et al., 2002), while the UK Digital Heart Project atlas captures population variation in ventricular shape and motion (Bai et al., 2015). Their modelling of coordinated variation across related cardiac structures is directly relevant to paired-surface reconstruction, where the two boundaries cannot be treated in isolation.

Shape-model studies have also identified variation associated with hypertrophic remodelling, normal anatomical diversity, and myocardial infarction (de Marvao et al., 2014; Medrano-Gracia et al., 2014; Suinesiaputra et al., 2018). Their findings show that clinically relevant anatomy is not captured by global volume alone. A population prior must therefore retain regional differences in proportion, apical form, and curvature rather than draw every case towards one mean geometry.

SSMs have two principal roles. They can constrain reconstruction to a learned shape space, or they can generate synthetic geometries by sampling that space (Heimann & Meinzer, 2009; Khalafvand et al., 2018). Synthetic sampling provides complete, plausible meshes that can increase the amount of geometric supervision available for learning-based methods.

Synthetic sampling increases the amount of complete geometric supervision, but it cannot introduce anatomy absent from the source atlas. It may also under-represent rare or severe pathology when the retained PCA modes favour common variation. Clinical data remain important for adapting a population prior to subject-specific and pathological anatomy while retaining the surface continuity learned from complete meshes. Combining SSM-derived and clinical data therefore assigns complementary roles to population-level priors and observed anatomical variation.

Before a learned model can be supervised with clinical data, the segmented SAX contours must be related to a three-dimensional surface. Linear contour interpolation, SSM fitting, and Radial Basis Function (RBF) fitting offer three different ways to supply the missing through-plane geometry. Their principal difference is the source of the reconstruction prior: linear interpolation uses only neighbouring contours, an SSM uses population-level modes of anatomical variation, and an RBF fit imposes smoothness through a continuous implicit field.

Linear contour lofting first establishes correspondence between points on adjacent rings and then joins those points with triangular strips. It is transparent, computationally inexpensive, and passes directly through the observed contours. However, the geometry between two slices is piecewise linear and depends on the chosen starting-point alignment and point correspondence. Large inter-slice gaps therefore produce abrupt changes in slope, while the apex and base require explicit closure rules. These limitations make lofting useful as a direct interpolation baseline, but less suitable as the sole source of smooth three-dimensional supervision from anisotropic SAX data (Petitjean & Dacher, 2011).

SSM fitting instead aligns a template with the observations and optimises its shape coefficients so that the deformed surface approaches the contours. The result retains a fixed topology and population-informed anatomical plausibility, which are valuable when observations are sparse (Cootes et al., 1995; Heimann & Meinzer, 2009). The same prior also limits the fit: anatomy outside the variation represented by the atlas cannot be recovered faithfully, and fitting requires a model appropriate to the structure and cardiac phase under study. An ED-only SSM cannot provide equivalent supervision for ES without imposing a diastolic shape space on systolic observations.

RBF reconstruction represents the surface as the zero level set of a smooth function fitted to scattered points (Carr et al., 2001). Surface samples are assigned zero values, while additional samples displaced to either side of the contour provide the sign needed to distinguish interior from exterior. The resulting implicit field can be evaluated on a regular grid. Marching Cubes then extracts a mesh from the sampled field (Lorensen & Cline, 1987). Unlike SSM fitting, this procedure does not require a phase-specific template or fixed mesh correspondence. Unlike linear lofting, it defines the intervening geometry as a smooth field rather than as strips between paired vertices. Its limitations are different: the reconstruction depends on the kernel, smoothing, and off-surface constraints, and strong regularisation can suppress genuine local variation.

2.3.2. Deep learning models for 3D shape reconstruction

Classical reconstruction methods depend on the sampled contours and on the assumptions imposed between slices, particularly near the apex and base, as discussed in Section 2.3.1. Learned reconstruction methods address the same trade-off between observation fidelity and geometric prior through two main families: template deformation and implicit field prediction.

Template methods learn to displace the vertices of a fixed mesh. Pixel2Mesh and Pose2Mesh demonstrate that graph-based deformation can recover dense geometry from sparse image or landmark evidence (Choi et al., 2020; N. Wang et al., 2018). In cardiac applications, fixed correspondence is valuable for population analysis and regional measurement (Bai et al., 2015; Beetz et al., 2022; Suinesiaputra et al., 2014). The trade-off is that the output remains tied to the topology and resolution of the chosen template. The same constraint that simplifies correspondence can therefore limit how far the reconstruction departs from the anatomy encoded by that template.

Implicit Neural Representations (INRs) remove the fixed output template by learning a continuous field from which a surface can be extracted at a chosen resolution. DeepSDF and Occupancy Networks established this approach using signed-distance and occupancy formulations, respectively (Mescheder et al., 2019; Park et al., 2019). For sparse cardiac reconstruction, removing the predefined mesh avoids restricting the output geometry to fixed resolution and connectivity. Output flexibility does not, however, guarantee that two predicted LV surfaces remain correctly ordered; anatomical validity must be addressed separately.

Recent cardiac work shows why implicit representations are useful when MRI slices are far apart. In a SAX scan, the detailed shape of the ventricle is observed within each slice, but less information is available between slices. Brás et al. (2026) addressed this anisotropy problem with a neural implicit representation that reconstructs the left and right ventricles, including the endocardial and epicardial boundaries of the LV myocardium. The study is closely related to sparse-contour reconstruction because both approaches seek continuous cardiac shape from incomplete observations. Br’as et al. use image data directly, whereas contour-based approaches operate on segmented point observations.

Muffoletto et al. (2026) also reconstruct three-dimensional cardiac shape from sparse segmentations with a neural implicit representation. Their method uses a learned biventricular shape prior and includes both SAX and long-axis views. It is therefore broader than LV-focused SDF reconstruction. The study does not report a public dataset or code release, which limits its direct reproducibility and comparison with other approaches.

The INR literature also provides mechanisms relevant to thin cardiac anatomy. Fourier features improve the recovery of local spatial variation, while Eikonal regularisation promotes a consistent distance field (Gropp et al., 2020; Tancik et al., 2020). In sparse-slice settings, both mechanisms can help mitigate the loss of local surface detail associated with interpolation across large gaps.

Recent cardiac INRs also model motion across the cardiac and respiratory cycles (Kunz et al., 2024; Quillien et al., 2025; Tian et al., 2026; R. Wang et al., 2025). Their main goal is temporally coherent image or volume reconstruction, whereas other implicit approaches focus specifically on continuous anatomical surface reconstruction conditioned on cardiac phase.

PointNet provides a complementary way to encode unordered observations (Qi et al., 2017). Its permutation-invariant formulation is relevant to SAX contours because arbitrary point ordering should not affect the inferred anatomy, making it suitable for point-based encoders in implicit surface models.

2.3.3. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. (1974) demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to end-diastolic wall thickness. More recently, Huelnhagen et al. (2017) showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, confirming its value as a non-invasive surrogate for myocardial tissue characterisation. The AHA 17-segment model (Cerqueira et al., 2002) provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a KD-tree identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. Its runtime depends on mesh size and the implementation. Nearest-neighbour distance can provide a lower bound on the true transmural thickness because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal until it intersects the epicardium. The intersection is computed via the M"oller–Trumbore algorithm, accelerated with a Bounding Volume Hierarchy (BVH) to $O(\log F)$ per query. Measuring along the surface normal better approximates a transmural direction but remains sensitive to normal noise: a slight rotation can produce a spurious long or short ray. The Shape Diameter Function variant improves robustness by casting a cone of K rays (typically $K{=}7$) at a half-angle of 30° around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxelised representations. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). The estimate is exact for planar parallel boundaries but can be biased in curved regions because the nearest-boundary directions do not necessarily follow a common transmural path. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm) but limited by voxel resolution.

Partial differential equation (PDE)-based methods are widely used for transmural thickness estimation. In a method developed for cortical rather than cardiac anatomy, Jones et al. (2000) solved Laplace's equation between the two tissue boundaries and used the resulting non-intersecting field lines to establish corresponding paths. The same geometric principle can be transferred to myocardial boundaries, with thickness obtained from the path length between the boundaries along these streamlines. Yezzi and Prince (2003) formulated an Eulerian PDE method that computes distances propagated from both boundaries along the Laplacian field. Their sum gives the local tissue thickness.

Finally, mesh-based regularised correspondence methods combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. Regularisation produces spatially smooth thickness maps suitable for bullseye visualisation, at the cost of potentially attenuating real local variation when the smoothing parameter is too aggressive.

2.3.4. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: deep learning gives accurate slice-wise segmentation, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. However, limited work integrates these components into a single pipeline for sparse SAX contour-based reconstruction.

A gap remains in methods that jointly address sparse SAX contour encoding, continuous paired endocardial and epicardial surface reconstruction, cardiac-phase conditioning, preservation of positive wall separation, and training with both clinical observations and SSM-derived synthetic geometries. Such an integrated framework would combine the complementary strengths of population-based shape priors, continuous implicit representations, and clinical contour data while supporting wall-thickness measurement beyond the acquired slices.

CHAPTER 3

Methodology

This chapter presents the methodology for reconstructing three-dimensional LV geometry from sparse SAX contours and measuring myocardial wall thickness on the resulting surfaces. The approach combines synthetic geometric supervision with clinical MRI-derived data, a phase-conditioned implicit representation, and a comparison of local wall-thickness estimators.

3.1. Overview

Sparse SAX contours constrain the LV only on the acquired image planes, leaving the intervening geometry to be inferred. Because clinical datasets do not provide independent three-dimensional surfaces for direct supervision, the training strategy first learns complete LV geometry from synthetic SSM samples and then adapts this prior to RBF-derived clinical meshes. Both stages use the same contour representation, allowing the model to receive a consistent form of sparse observation throughout training.

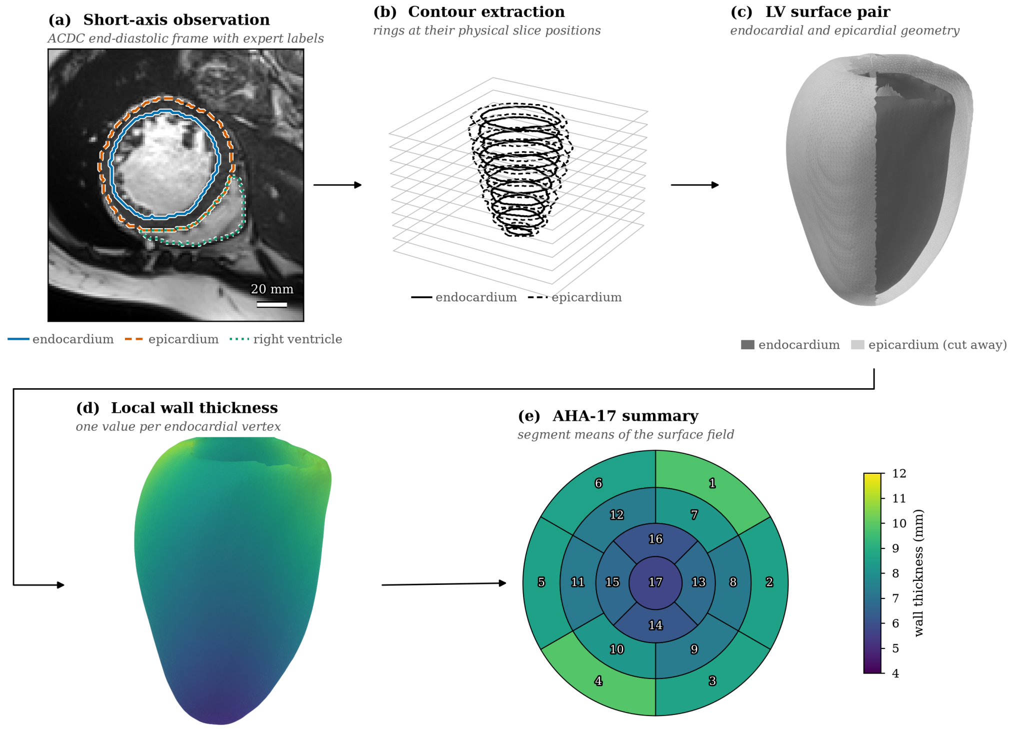

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Figure 3.1. Methodology from sparse SAX observations to reconstructed LV surfaces and wall-thickness outputs. ACDC data are shown in panels (a)–(b); panels (c)–(e) use illustrative SSM-derived geometry rather than reported results.

"‘‘

Reconstructing the endocardium and epicardium as unrelated surfaces could produce intersections or local wall collapse. The model therefore uses a phase-conditioned implicit decoder that reconstructs the two boundaries with an explicit positive separation. After surface extraction, several local wall-thickness estimators are compared because the resulting measurement depends not only on the reconstructed geometry, but also on how correspondence across the myocardial wall is defined. Regional AHA-17 aggregation is performed only after the local measurement method has been selected.

Figure 3.1 summarises the complete workflow. The synthetic epicardial offset shown in panels (c)–(e) is used only to illustrate the geometric reconstruction and measurement stages and does not represent a clinical wall-thickness reference.

The chapter follows this dependency. Section 3.2 establishes the synthetic and clinical supervision, Section 3.3 defines the contour encoder and coupled decoder, and Section 3.4 describes model optimisation and surface extraction. Section 3.5 then evaluates the local thickness estimators and regional aggregation.

3.2. Data Preparation

A fundamental challenge in training the model is the absence of a paired dataset that provides both sparse SAX contour observations *and* independently acquired three-dimensional LV meshes. Clinical cardiac MRI datasets supply segmentation masks and image volumes, but they do not include the watertight endocardial and epicardial surfaces that an SDF decoder requires as supervision. This absence of independent three-dimensional targets is the starting point for the entire data-preparation strategy.

The training data were therefore constructed in two stages. First, 1300 synthetic ED geometries were generated from the UK Digital Heart Project LV SSM for pre-training. These samples provide controlled variation consistent with the shape distribution represented by the SSM. Second, ED and ES segmentations from ACDC and M&Ms-2 were converted into watertight meshes by fitting implicit Radial Basis Function (RBF) surfaces to their SAX contours. These cases introduce patient-specific anatomy, acquisition variability, and cardiac-phase variation during fine-tuning.

Both stages use a common cache schema, which keeps the model interface and loss formulation unchanged as training moves from synthetic to clinical data. The schema also separates the sparse contours supplied to the encoder from the mesh-derived supervision used only during training.

3.2.1. Synthetic supervision and SSM pre-training

The synthetic and clinical streams address different parts of the supervision gap. Sampling the SSM supplies complete, watertight surfaces over a controlled range of LV geometry, but it cannot reproduce scanner-dependent acquisition, clinical label variation, or systolic anatomy. Conversely, the ACDC (Bernard et al., 2018) and M&Ms-2 (Martín-Isla et al., 2023) labels contain patient-specific ED and ES observations, but their slice-wise segmentations must be converted into closed surfaces before they can supervise the implicit decoder. Synthetic pre-training and clinical fine-tuning are used in that order so that the model first learns a regularised geometric prior and then encounters the conditions present in the evaluation data.

Fitting the SSM directly to the clinical contours was considered for this conversion. When the data-preparation pipeline was developed, however, the available UK Digital Heart Project model represented

ED but not ES. Using it for both phases would have constrained systolic cases to a diastolic shape space, whereas using it only for ED would have introduced a phase-dependent difference in how the clinical targets were constructed. RBF fitting was therefore adopted for both phases. This preserves one reconstruction procedure across ACDC and M&Ms-2 while leaving the SSM in its intended role as the source of synthetic ED pre-training geometry.

The synthetic pre-training data were generated from the UK Digital Heart Project LV SSM reported by Bai et al. (2015), built from more than 1,000 high-resolution cardiac MR images of healthy subjects. The model provides a mean endocardial surface of 22 043 vertices together with 100 principal modes of variation stored as a $66\,129 \times 100$ matrix of mode vectors and a corresponding vector of 100 eigenvalues λ_i. Following the standard SSM formulation (Heimann & Meinzer, 2009), a new endocardial mesh is generated by

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape flattened to a vector, Φ is the matrix of mode vectors, and $b = [b_1, \ldots, b_{100}]^\top$ is the sampled coefficient vector. The standard deviation $\sigma_i = \sqrt{\lambda_i}$ sets the scale of variation along mode i.

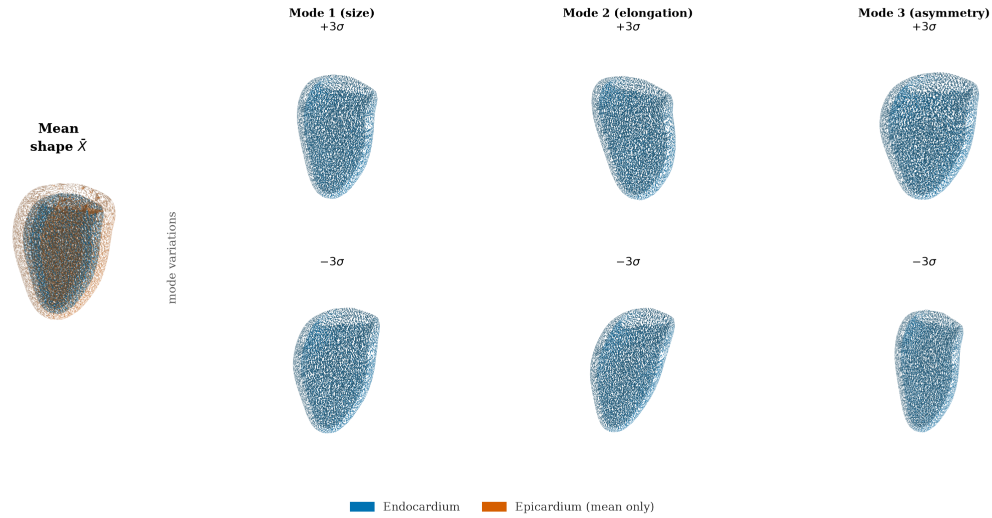

Figure 3.2. UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$.

The first few modes capture the largest sources of shape variation in the population; mode 1 primarily controls overall LV size, mode 2 encodes elongation along the long axis, and higher modes represent progressively more localised deformations, as shown in Figure 3.2.

The synthetic ED dataset was built by sampling 1300 meshes from the SSM and converting each mesh into a training-compatible sample. The generation pipeline combines latent sampling, mesh generation, quality filtering, epicardial construction, and supervision extraction, as illustrated in Figure 3.3.

Each candidate coefficient vector b was drawn by sampling each component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and then applying two acceptance criteria:

$$\left|\frac{b_i}{\sigma_i}\right| \leq 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \leq \chi^2_{0.99,d}, \tag{3.2}$$

where σ_i is the standard deviation of mode i, d is the number of active modes, and $\chi^2_{0.99,d}$ is the 99th percentile of the chi-squared distribution with d degrees of freedom. The per-mode clipping prevents individual modes from dominating the shape, and the Mahalanobis bound prevents the combined deformation from falling outside the plausible population distribution. Accepted samples are scattered within a bounded ellipsoid in latent space, as shown in panel (c) of Figure 3.3.

Each accepted coefficient vector b is substituted into Equation (3.1) to produce a deformed endocardial vertex array, which is then triangulated using the face connectivity of the mean-shape mesh. The resulting triangular surface is watertight and oriented by construction.

Before a generated mesh is accepted into the training set, it is evaluated against four proxy metrics: enclosed volume (mL), surface area (cm^2), sphericity index, and plausibility of the apical and basal normal directions. Samples that fall outside calibrated bounds for any of these metrics are discarded. Periodic exact audits using VTK mass-properties computation were also performed every 40 samples to verify the proxy metric calibration.

The SSM provides only the endocardial surface. The epicardial surface was synthesised by offsetting each endocardial vertex p_i along its outward normal n_i by a spatially varying distance d_i,

$$q_i = p_i + d_i n_i, \tag{3.3}$$

where d_i was set to approximately 10 mm near the base and 5 mm near the apex, with additive Gaussian noise of standard deviation 0.5 mm to avoid perfectly uniform geometry. This synthetic epicardium is a geometric pre-training prior only. It is not intended to reproduce clinical wall thickness, and it is never used as a reference for any wall-thickness result. Its purpose is to give the decoder a consistent outer surface during pre-training, so that the model learns the general shape relation between an inner and an outer LV surface before it sees real data.

Each pair of endocardial and epicardial surfaces was sliced at 10 evenly spaced axial levels to produce sparse SAX contour rings, matching the acquisition pattern of real cine MRI. For each case, 2048 query points were sampled near the surfaces and in the surrounding volume, and signed-distance targets or inside/outside labels were computed. These query points form the dense volumetric supervision used by the SDF loss terms during training.

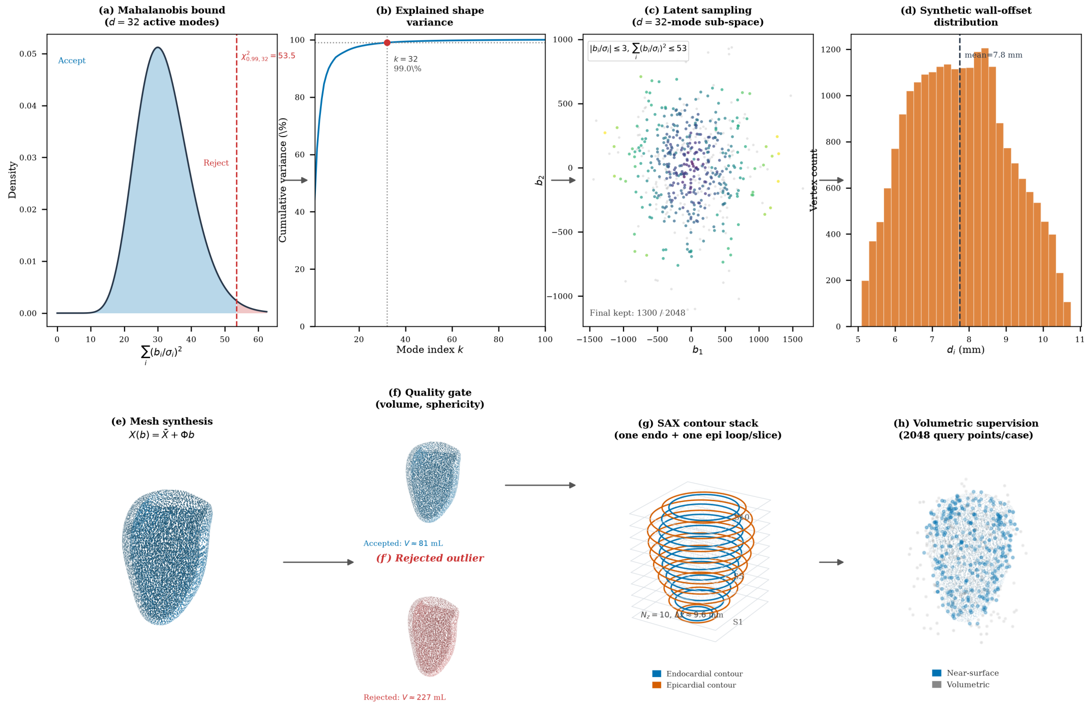

Figure 3.3. Synthetic ED dataset generation from SSM sampling to geometric supervision. Top row: statistical constraints; bottom row: geometric construction and supervision targets.

Figure 3.3 summarises this process in two rows. The top row provides the statistical view of sampling: panel (a) shows the Mahalanobis-statistic density and the $\chi^2_{0.99,32}$ acceptance threshold, panel (b) shows that 32 active modes capture 99% cumulative shape variance, panel (c) shows accepted latent samples in the first two dimensions, and panel (d) shows the synthetic epicardial-offset distribution from Equation (3.3). The bottom row provides the geometric view: panel (e) shows mesh synthesis via Equation (3.1), panel (f) contrasts an accepted shape with a rejected outlier, panel (g) shows the 10 SAX contour levels, and panel (h) shows the 2048 near-surface and volumetric query points used for decoder supervision.

These fixed SSM-sampling and supervision settings were kept constant across all synthetic ED cases to ensure reproducible pre-training before fine-tuning on real ED and ES data.

3.2.2. Clinical ED and ES supervision from SAX labels

SSM-derived meshes alone cannot capture the full diversity of clinical acquisition conditions: scanner differences, segmentation quality, image resolution, and the distinct geometry of the end-systolic phase. A second data stream was therefore created by reconstructing watertight three-dimensional LV meshes directly from the segmentation masks provided in the ACDC (Bernard et al., 2018) and M&Ms-2 datasets, taking both the ED and ES frames so that the model is exposed to both cardiac phases.

These real ED and ES caches are used in the second optimisation stage, after the initial SSM-based training. The reason for this fine-tuning stage is that the synthetic corpus teaches a strong global LV-shape prior, but it does not fully reproduce clinical variability in contour quality, inter-dataset label style, and phase-specific contraction patterns. Fine-tuning on real ED and ES therefore reduces domain shift between synthetic supervision and the target clinical setting while preserving the geometric regularity learned from the SSM.

Operationally, the fine-tuning stage keeps the same model architecture and the same multi-term loss from Section 3.4, but replaces the synthetic-only stream with RBF-derived cache samples from ED and ES frames. Because the real caches follow the same schema (contours, tissue labels, phase label, mesh supervision, and query points), the training loop can continue from the pre-trained checkpoint without changing model interfaces. The phase label then provides the conditioning signal that allows one shared model to adapt from diastolic to systolic geometry during continued training.

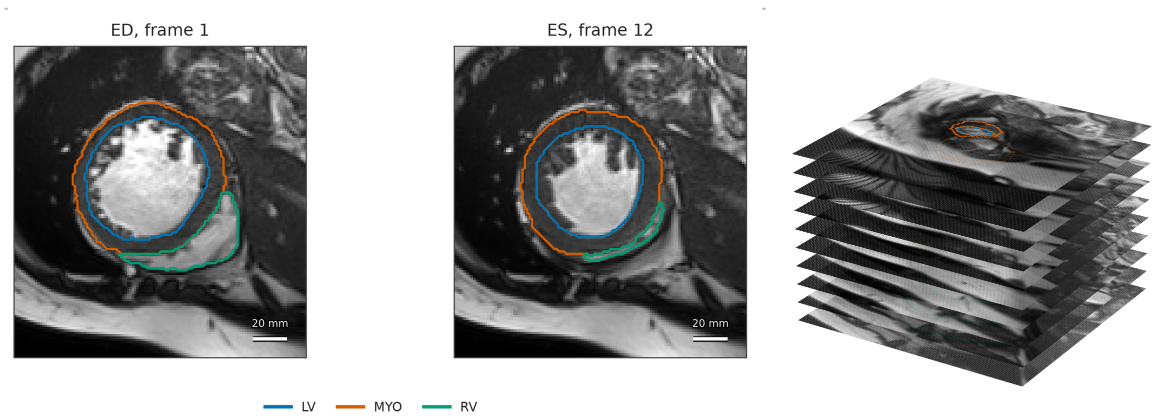

Figure 3.4. Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding sparse ED SAX stack in physical three-dimensional spacing.

Before describing how these meshes are built, it is useful to look at the raw clinical input, because it makes clear why a reconstruction step is needed at all. Figure 3.4 shows representative SAX slices from the ACDC dataset (Bernard et al., 2018), with the LV cavity, myocardium, and right-ventricular contours overlaid. Each frame is only a two-dimensional cross-section of the heart, and the segmentation is defined one slice at a time. Panel (b) of Figure 3.4 stacks the segmented ED slices at their physical through-plane positions and exposes the central difficulty of the problem: a clinical acquisition provides only a small number of parallel SAX slices separated by several millimetres, so the observed geometry is a set of sparse contour rings rather than a continuous surface.

The reconstruction pipeline closes this gap by fitting a continuous implicit surface directly to the sparse contours of each clinical label stack. The resulting meshes are *RBF-derived surrogate targets*, not manually annotated or independently acquired 3D reference geometry. Their shape is determined by the observed contour points and by the RBF kernel, off-surface constraints, field-domain regularisation, and extraction resolution described below. This distinction is important when interpreting the reconstruction-consistency results.

Each NIfTI segmentation is reoriented to the RAS+ convention while preserving its physical voxel spacing. Dataset-specific labels are mapped to a common definition of the LV cavity and myocardium. On every acquired SAX slice, the largest endocardial and epicardial rings are retained, where the endocardium is the boundary of the LV cavity and the epicardium is the outer boundary of the union of cavity and myocardium. Each ring is resampled to 60 evenly spaced points and stored at its measured through-plane position. These points are both the observations supplied to the encoder and the constraints from which the clinical surrogate surfaces are constructed.

The endocardial and epicardial surfaces are fitted separately. For either surface s, let p_i be a contour point and n_i its outward in-plane normal, estimated from the local contour tangent and oriented away from the ring centre. The implicit function is a thin-plate-spline RBF with a linear polynomial term,

$$f_s(x) = \sum_{j=1}^{K} w_j \phi(\|x - c_j\|_2) + a_0 + a^\top x, \qquad \phi(r) = r^2 \log r, \tag{3.4}$$

following the implicit surface formulation of Carr et al. (2001). The contour points constrain the zero level set, while paired off-surface points at $d = 2.5$ mm establish its sign:

$$f_s(p_i) = 0, \qquad f_s(p_i + dn_i) = d, \qquad f_s(p_i - dn_i) = -d. \tag{3.5}$$

A regularisation parameter of 0.5 reduces sensitivity to small contour irregularities during RBF fitting. Unlike direct lofting, the fit does not require a vertex on one ring to be paired with a particular vertex on the next. Unlike SSM fitting, it does not constrain either phase to an atlas-derived shape space.

Each fitted field is evaluated on a 1.0 mm isotropic grid. The grid extends 8.0 mm beyond the contours in-plane and 1.5 mm beyond the first and last rings along the SAX direction, so that the extracted surface closes close to the observed longitudinal support. The interior defined by $f_s \leq 0$ is reduced to its valid connected region and converted to a signed-distance volume. A Gaussian filter with a physical standard deviation of 1.2 mm is applied to this field before marching cubes extracts the zero level set (Lorensen & Cline, 1987). Regularising the field before extraction reduces voxel-scale corrugation without directly displacing mesh vertices after the surface has been formed. The extracted mesh is then checked for a single closed component, consistent face orientation, and genus-zero topology. When these conditions hold, the geometry is preserved unchanged. Otherwise, component selection, hole closure, orientation correction, and genus normalisation are applied conditionally. Cases with fewer than two valid rings per surface, failed surface extraction, unresolved topology, implausible dimensions, or invalid endocardial–epicardial containment are excluded before caching.

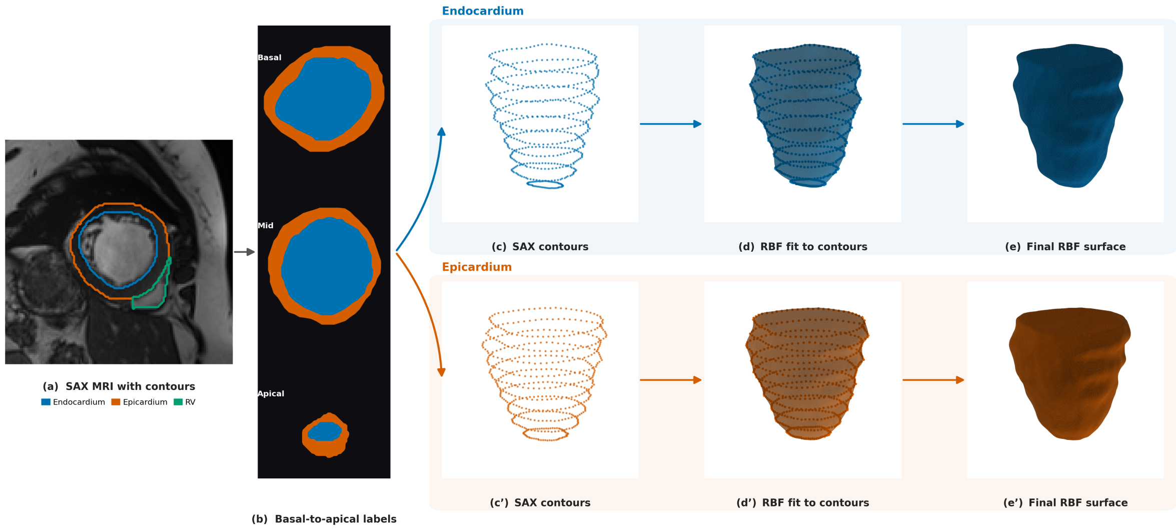

Figure 3.5. Clinical RBF contour-to-surface pipeline. Panel (a) shows one acquired SAX slice with the expert endocardial, epicardial, and right-ventricular contours. Panel (b) shows the corresponding basal, mid-cavity, and apical labels, while panels (c) and (c′) stack the extracted contour rings. Panels (d) and (d′) overlay those constraints on the zero-level surfaces extracted from the regularised RBF fields, and panels (e) and (e′) show the final topology-validated surfaces used to construct the clinical supervision.

Figure 3.5 summarises the implemented clinical-data pipeline. The two branches show that the endocardial and epicardial constraints are fitted separately before their extracted surfaces undergo the same topology checks. Valid surfaces pass through this stage unchanged; topology correction is invoked only when a surface fails those checks.

The RBF-derived endocardial and epicardial meshes are centred and normalised with the same translation and scale. For each pair, 2048 query points are drawn near the surfaces and throughout a padded bounding box. Near-surface points constrain the location of each boundary, while volumetric points determine the inside/outside relation. The contours remain the sparse model input; the mesh samples, query points, and their targets are available only as supervision.

3.2.3. Shared dataset assembly and partitioning

The synthetic and clinical streams are assembled through one cache contract, one augmentation protocol, and patient-level partitions that define their roles in training and evaluation.

All cases are stored in a common cache representation so that the model receives the same interface regardless of whether a sample originates from the SSM or from a clinical segmentation.

A cache item combines the sparse contour observation with mesh-derived supervision and the metadata needed to keep both representations in a shared coordinate frame. The ED and ES caches contain 10 SAX levels, 60 points per available contour ring, and 2048 query points per case. This design deliberately separates what the model observes from what it is allowed to learn from: the encoder receives only contour coordinates, tissue identity, and cardiac phase, whereas the decoder is supervised by surface and volumetric samples derived from the corresponding mesh. Thus, at inference time, the same model can reconstruct geometry from sparse contour observations alone.

Table 3.1. Fields of one shared cache item and their role in training.

Field	Origin	Role
Contour points	Synthetic mesh slicing or clinical slice labels	Sparse SAX observation for the encoder and contour-anchor loss
Tissue label	Contour extraction	Distinguishes endocardial from epicardial points during tissue-specific pooling
Phase label	Dataset metadata	Conditions the encoder on ED or ES geometry
Surface samples	Prepared endocardial and epicardial meshes	Surface-fitting and normal-alignment supervision
Query points and targets	Near-surface and volumetric mesh sampling	Dense SDF, occupancy, sign, and exterior supervision
Mesh metadata	Preprocessing pipeline	Coordinates, scale, faces, and orientation for evaluation and visualisation

Table 3.1 makes this separation explicit. The first three fields describe the clinical-style observation supplied to the encoder; the remaining fields are available only during training, evaluation, or visualisation. Keeping these roles distinct prevents mesh-derived information from leaking into the encoder input while allowing dense supervision of the implicit field.

This common contract enables two-stage optimisation without dataset-specific changes to the architecture or loss. The model is first trained on synthetic ED samples, which establish a broad geometric prior, and is then fine-tuned on real ED and ES samples, which introduce patient-specific anatomy and phase-dependent shape variation. The loader changes only the source stream between stages; the fields consumed by the training loop remain unchanged.

Contour augmentation is applied to both synthetic ED and real ED/ES training observations, after the train/validation/test split. The target mesh, surface samples, query points, occupancy labels, tissue labels, and phase label remain fixed. This is therefore an observation-robustness intervention rather than the creation of a geometrically transformed anatomical target. Such label-preserving perturbations are commonly used to improve robustness when labelled data are limited (Chlap et al., 2021; Shorten & Khoshgoftaar, 2019).

For each augmented sample, the loader draws small independent in-plane translations per slice, point-wise in-plane jitter, global scale jitter, and contour-point dropout. Slice dropout and long-axis rotation are applied probabilistically while retaining at least three slice levels. The endocardial and epicardial contours within a slice receive the same translation, preserving their local relationship. As illustrated in Figure 3.6, these perturbations ask the encoder to recover one fixed mesh-derived target from plausible variations of its sparse SAX observation.

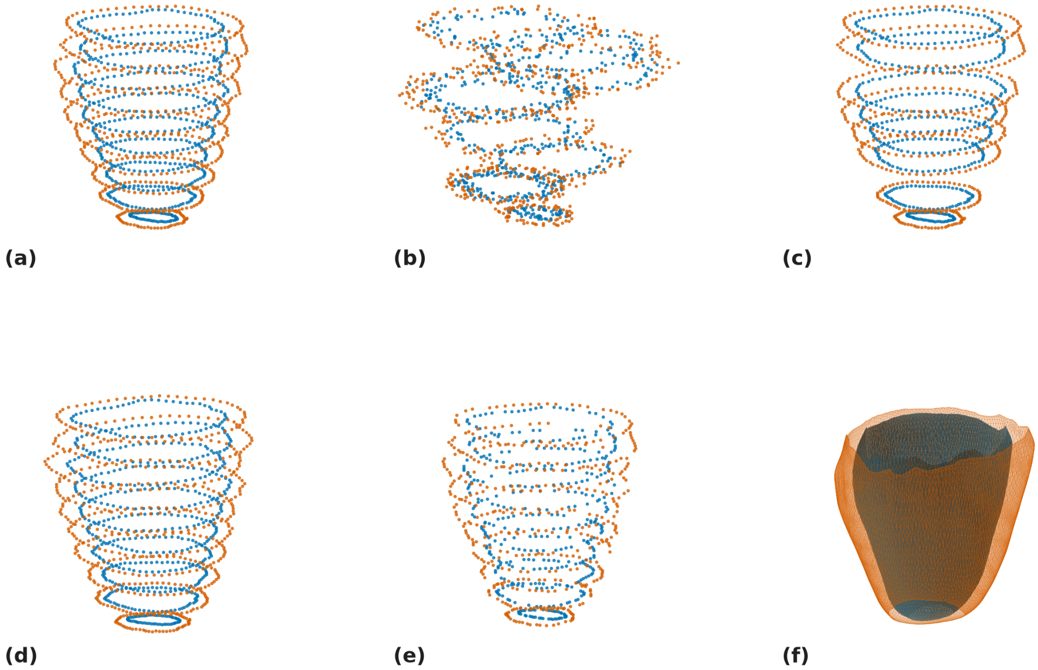

Figure 3.6. Examples of contour-input augmentation. Endocardial and epicardial points are shown in blue and orange, respectively.

In Figure 3.6, panel (a) shows the unaltered contour stack, while panels (b), (d), and (e) show slice translation, slice rotation, and contour-point dropout, respectively. Panel (f) shows the corresponding 3D mesh label. Thus, the sparse SAX observation changes across the augmented examples, whereas the derived mesh and all decoder-supervision targets remain fixed; the figure illustrates robustness of the encoder input rather than transformation of the anatomical ground-truth geometry.

This subsection reports how many samples each data stream finally contributes and how the real streams are divided for training and evaluation. The real data come from two public collections: the Automated Cardiac Diagnosis Challenge (ACDC) and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (M&Ms-2). Together they provide 460 patients (100 from ACDC and 360 from M&Ms-2). For each patient we reconstruct one ED mesh and one ES mesh, giving two real streams. Both collections are openly available for research at no cost; each requires a free registration and the acceptance of a data-use agreement. The M&Ms-2 dataset is documented by Martín-Isla et al. (2023) and contains the 360 patients used here. The synthetic stream is generated from the UK Digital Heart Project SSM: of its 100 modes, the first 32 already capture 99% of the shape variance, so we sample within that sub-space. Drawing shape coefficients under the Mahalanobis bound of Equation (3.2) and rejecting implausible shapes yields 1300 accepted meshes out of 2048 candidates (63%). The shape model is likewise open and free: it is distributed as a public repository by the UK Digital Heart Project,[1] needs no access application, and the authors ask only that the atlas papers be cited. None of the three sources used in this work therefore required a licence fee.

1

https://github.com/UK-Digital-Heart-Project/Statistical-Shape-Model

Not every real patient produces a usable mesh. A segmentation may be a long-axis view, cover too few slices, or fail an anatomical plausibility check, such as an endocardium that is not fully enclosed by the epicardium. After these quality gates, 444 of the 460 patients yield a valid ED mesh and 423 yield a valid ES mesh. The surviving meshes are then divided for training and evaluation. To prevent information from leaking between the partitions, the real streams are split by patient rather than by mesh, so that all meshes from a given patient stay within a single partition. Both streams use the same 70 %/15 %/15 % allocation for training, validation, and test, rounded to whole patients, and the synthetic ED meshes are used only to enrich the pre-training stage and are never placed in the validation or test sets.

Table 3.2. Final sample counts and partitioning of the three data streams. Real streams report patients that pass the quality gates; the synthetic stream reports meshes accepted out of the 2048 sampled shape vectors. The reported clinical evaluation contains no synthetic cases.

Stream	Source	Phase	Count
Real data	ACDC and M&Ms-2	ED	444 patients
Real data	ACDC and M&Ms-2	ES	423 patients
Synthetic data	UK Digital Heart SSM	ED	1300 meshes

Table 3.2 records the two real source collections and the synthetic pre-training stream. The role of each stream in the two-stage training procedure is summarised in Appendix A.

3.3. Model Architecture

The proposed model maps sparse endocardial and epicardial SAX contours to two continuous implicit fields. Its architecture follows the information required at each stage of this mapping. A point-based global encoder first summarises the overall ventricular geometry in a compact latent vector. In parallel, a local encoder scatters contour features into a coarse three-dimensional volume, thereby retaining where the observed wall lies in space. An implicit decoder then combines the global description, the local feature sampled near each query point, and a positional encoding of that point to predict the endocardial field and its separation from the epicardial field.

The global and local pathways are complementary. Global max-pooling makes the representation independent of contour-point order and captures the chamber-level shape observed across all slices, but compressing the complete case into one vector removes explicit spatial correspondence. The local feature volume preserves this correspondence at a coarse resolution and allows nearby contour evidence to influence individual field queries. Together, the two pathways support geometric completion between slices without restricting the output to the topology, resolution, or vertex count of a template mesh.

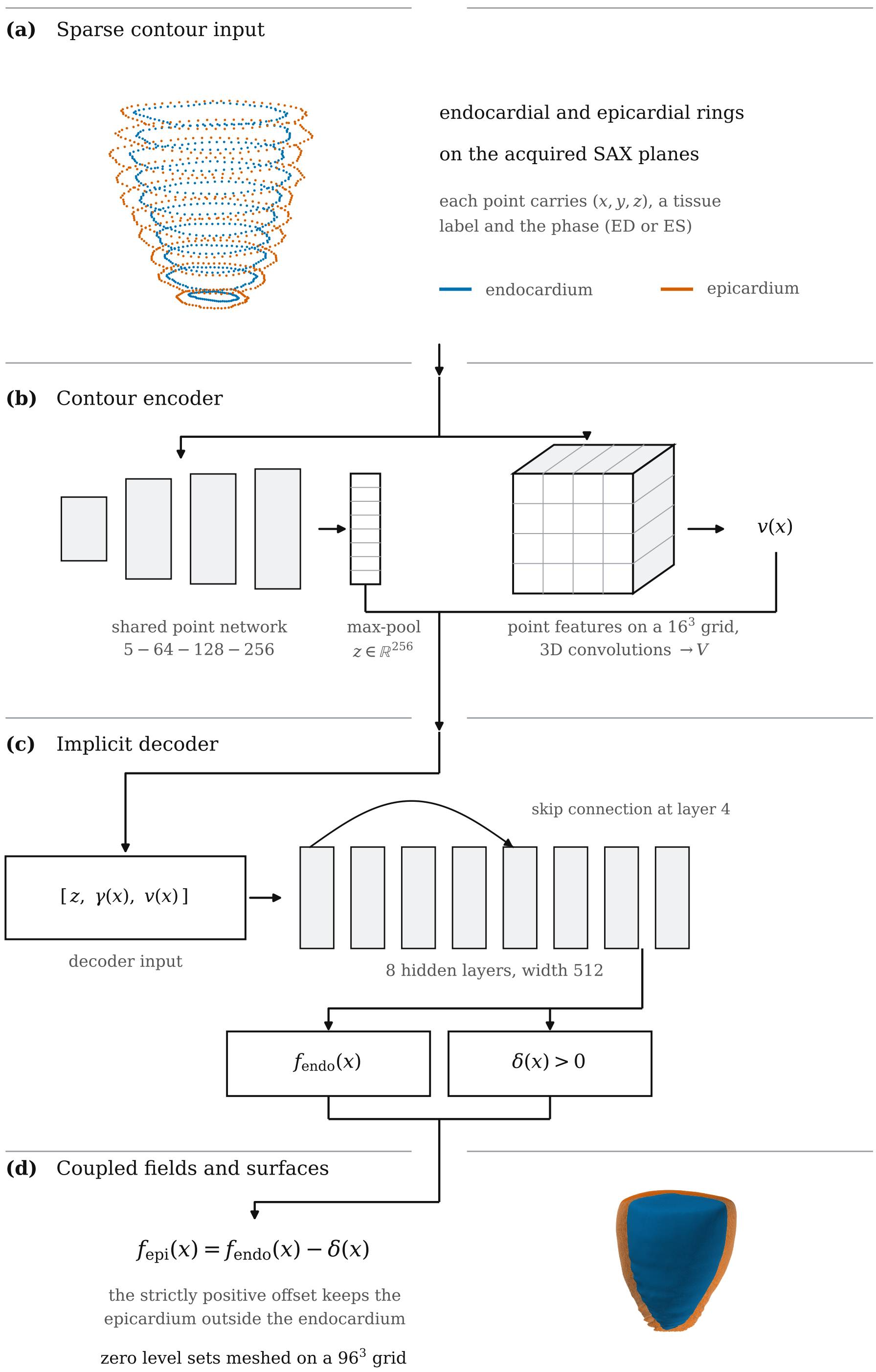

Figure 3.7. Phase-conditioned model architecture, from the sparse contour observation (a) through the global and local encoder paths (b) and the implicit decoder (c) to the coupled fields and extracted surfaces (d). The point cloud in (a) and the cut surfaces in (d) are the observed contours and the corresponding reconstruction of one ACDC case, with the epicardium opened to expose the endocardial cavity.

Figure 3.7 links the global shape code, local feature volume, and coupled decoder. The following subsections describe the input representation, the two encoder pathways, and the positive coupling of the output fields. The offset shown in the figure is a separation between implicit fields, not a measured transmural distance.

3.3.1. Input and conditioning representation

For a batch of cases, the encoder receives a padded contour tensor

$$P \in \mathbb{R}^{B\times N\times 5} \tag{3.6}$$

and a validity mask $M \in \{0,1\}^{B\times N}$. Each valid row is $p_i = (x_i, y_i, z_i, t_i, q_i)$, where (x_i, y_i, z_i) are normalised spatial coordinates, t_i distinguishes endocardial from epicardial points, and q_i identifies ED or ES. Padding therefore permits different numbers of observed points within one batch without treating padded positions as anatomy. The network receives neither image intensities nor a template mesh: all case-specific conditioning is derived from the sparse contour coordinates, tissue identity, and cardiac phase.

The phase value is attached to every point rather than introduced only at the decoder. Consequently, both the global and local encoders can interpret the same contour arrangement differently for ED and ES. Tissue identity is retained for a different purpose: it separates evidence about the cavity boundary from evidence about the outer myocardial boundary before both are combined into a shared case representation.

3.3.2. Contour encoder

The encoder contains separate global and local pathways. The global pathway uses a shared point network with dimensions $5 \to 64 \to 128 \to 256 \to 256$. Because the same weights process every point independently, no ordering is imposed within a contour or between slices. ReLU activations follow the intermediate linear layers, producing a 256-dimensional feature h_i for each valid point.

More formally, let $P = \{p_i\}_{i=1}^{N}$ be the contour points of one case, where each p_i carries coordinates, a tissue label, and a phase label. The shared point network computes

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.7}$$

where ϕ_{enc} is the shared network. The endocardial and epicardial descriptors are then obtained by max-pooling within each tissue,

$$z_{\text{endo}} = \max_{i\in\mathcal{E}} h_i, \qquad z_{\text{epi}} = \max_{i\in\mathcal{P}} h_i, \tag{3.8}$$

where $\mathcal{E}$ and $\mathcal{P}$ are the endocardial and epicardial point sets. The maximum is taken independently in every feature channel. It therefore retains the strongest response to each learned geometric pattern while remaining invariant to permutations of the input points.

The two tissue descriptors are concatenated into a 512-dimensional vector and projected to the global latent code

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z, \tag{3.9}$$

where $z \in \mathbb{R}^{256}$. Separate pooling prevents evidence from the inner and outer contours from being merged before the network has formed a descriptor for each boundary. Their projection into one vector then provides the decoder with a compact description of the complete case.

The local pathway addresses the spatial detail that global max-pooling cannot retain. A second shared network, with dimensions $5 \to 64 \to 64 \to 32$, assigns a 32-dimensional feature to every valid contour point. These features are placed in a regular 16^3 grid covering the normalised extent $[-1.8, 1.8]^3$. When several points occupy the same voxel, their features are averaged, so the volume does not depend directly on the number of contour samples at that location. Three $3 \times 3 \times 3$ convolutions, with Group Normalisation and ReLU after the first two, propagate information into neighbouring voxels and produce

$$V \in \mathbb{R}^{32\times16\times16\times16}. \tag{3.10}$$

For a query point x, the surrounding local feature is obtained by trilinear interpolation,

$$v(x) = \mathsf{Interp}(V, x). \tag{3.11}$$

The resulting vector $v(x) \in \mathbb{R}^{32}$ describes contour evidence near the query location, whereas z remains constant for every query belonging to the same case. The local grid is deliberately coarse: it supplies spatial context to the decoder rather than serving as the final reconstruction grid.

3.3.3. Implicit decoder and positive-wall coupling

The decoder represents the surfaces as functions of continuous coordinates. Before a query point is processed, its three coordinates are expanded with Fourier features, which provide periodic components at several spatial frequencies and improve the representation of geometric variation (Tancik et al., 2020). For $x \in \mathbb{R}^3$, the encoding is

$$\gamma(x) = [x, \sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)], \tag{3.12}$$

with $L = 3$. The encoded coordinate therefore has $3 + 6L = 21$ components. It is concatenated with the global latent vector to form

$$h_{\mathsf{in}}(x) = [z, \gamma(x)] \in \mathbb{R}^{277}. \tag{3.13}$$

An input projection maps this vector to width 512, after which the decoder trunk applies eight fully connected hidden transformations of the same width. Softplus activations provide differentiable transitions throughout the field. At hidden layer 4, the original 277-dimensional decoder input is concatenated with the current state before projection back to width 512. This skip connection gives the deeper layers renewed access to the case-level and coordinate information instead of requiring both to pass unchanged through the entire trunk.

The sampled local feature is injected into the decoder at its input and skip connection, allowing the prediction at each coordinate to depend on nearby contour evidence as well as on the global ventricular shape. The final hidden representation feeds two coupled output paths. One predicts the endocardial signed-distance value, while the other predicts a positive field offset δ. The epicardial field is then defined by

$$f_{\mathsf{epi}}(x) = f_{\mathsf{endo}}(x) - \delta(x), \qquad \delta(x) > 0. \tag{3.14}$$

The implementation enforces positivity through a sigmoid-based mapping with additional headroom for larger offsets. Defining the epicardial field from the endocardial prediction reduces the risk of local wall collapse compared with two independent output fields. However, the constraint does not by itself guarantee anatomically valid surfaces or valid mesh topology after discretisation. Furthermore, δ is a difference between implicit field values, not a Euclidean correspondence between the extracted surfaces, and must therefore not be interpreted directly as anatomical wall thickness.

During inference, z and V are computed once from the contour set, after which the decoder can be evaluated at arbitrary coordinates. The implementation samples both fields on a dense 96^3 grid and applies marching cubes separately to their zero-level sets. Mesh generation is therefore an inference-time discretisation of the continuous representation, not an architectural restriction on the number or connectivity of output vertices. The post-processing and measurement steps applied to these meshes are described in Section 3.4.2.

The complete architecture configuration is collected in Table A.3 of Appendix A.

3.4. Training and Inference

This section explains how the model is trained from the shared cache and how a surface mesh, decoder offset map, and subsequent wall-thickness measurements are produced once training is finished. Training proceeds in two stages: the model is first pre-trained on the synthetic SSM cases and then fine-tuned on the real ED and ES cases. In both stages it must learn two things together: a valid distance field for the endocardium, and a sensible relationship between the endocardium and the epicardium.

3.4.1. Model training and optimisation

Training follows the two-stage strategy introduced in Section 3.2. In the first stage, the model is pre-trained only on the synthetic ED cases generated from the SSM. This controlled corpus provides the model with a broad geometric prior for LV shape and allows stronger augmentation of the sparse observations.

In the second stage, the pre-trained model is fine-tuned on the real cases reconstructed from ACDC and M&Ms-2, covering both the ED and ES phases. Fine-tuning starts from the pre-trained weights and continues with the same architecture and loss, so the model keeps the geometric regularity learned from the SSM while adapting to real segmentation quality and to the systolic phase. Throughout both stages the phase label is provided as input, so a single model handles ED and ES. Contour augmentation is used for both synthetic ED and real ED/ES observations; their surrogate targets and cached supervision are never changed.

A single phase-conditioned model is used instead of two separate models. When the experiments were designed, the available SSM provided synthetic geometry only for ED, so a separate ES model could not receive equivalent synthetic pre-training. Sharing one model allows the ES cases to benefit from the broad shape prior learned from the synthetic ED corpus, while the phase label tells the encoder which configuration it is processing. The RBF-derived clinical targets use the same construction procedure in both phases and therefore avoid introducing a second asymmetry at the fine-tuning stage. The cost is that phase information remains coarse: the model receives a binary label rather than a continuous position in the cardiac cycle. Since the synthetic corpus is ED-only, the prior also remains biased towards diastolic shape.

The two stages are complementary. The synthetic ED cases give a large set of anatomically valid shapes but do not reproduce every artefact seen in real segmentations, which is why they are used for pre-training rather than as the final training data. Their role is to provide a geometric prior: they regularise the shape space, expose the model to a wide range of plausible LV geometries, and do so under controlled conditions. The RBF-derived real ED and ES cases then supply patient-specific anatomy from several datasets and let the model learn how the shape changes between the two phases. To avoid information leaking between the training, validation, and test sets, the real cases are split by patient before augmentation. Because augmentation perturbs only the observed contours, each real surrogate target remains tied to the anatomy of its own patient.

In practice, this means the encoder is trained to be robust to imperfect, noisy inputs, such as small shifts, jitter, missing slices, or dropped points, while the decoder is always supervised against a fixed, consistent 3D geometry.

The training objective contains five terms. Let $\mathcal{E}$ and $\mathcal{P}$ denote sampled endocardial and epicardial surface points, $\mathcal{Q}$ the cached SDF query points, $\mathcal{R}$ the union of near-surface and free-space points used for gradient regularisation, and $\mathcal{F}$ the free-space subset. At each $q \in \mathcal{Q}$, $y_{\text{endo}}(q)$ and $y_{\text{epi}}(q)$ are the cached signed-distance targets. At each $x \in \mathcal{E}$, the wall target is $t(x) = -y_{\text{epi}}(x)$, the unsigned epicardial distance evaluated on the endocardial surface. The implemented loss terms are

$$
\begin{aligned}
\mathcal{L}_{\text{surf}} &= \mathbb{E}_{x\in\mathcal{E}}[|f_{\text{endo}}(x)|] + \mathbb{E}_{x\in\mathcal{P}}[|f_{\text{epi}}(x)|], \\
\mathcal{L}_{\text{SDF}} &= \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{endo}}(q) - y_{\text{endo}}(q)|] + \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{epi}}(q) - y_{\text{epi}}(q)|], \\
\mathcal{L}_{\text{eik}} &= \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{endo}}(x)\|_2 - 1)^2] + \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{epi}}(x)\|_2 - 1)^2], \\
\mathcal{L}_{\text{off}} &= \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{endo}}(x)|}] + \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{epi}}(x)|}], \\
\mathcal{L}_{\text{WT}} &= \mathbb{E}_{x\in\mathcal{E}}[\ell_{\beta_{\text{WT}}}(\delta(x) - t(x))],
\end{aligned} \tag{3.15}
$$

where $\alpha = 5$ and ℓ_β is the Smooth-L1 penalty

$$
\ell_\beta(a) = \begin{cases} a^2/(2\beta), & |a| < \beta, \\ |a| - \beta/2, & |a| \geq \beta, \end{cases} \qquad \beta_{\text{WT}} = 0.02. \tag{3.16}
$$

All distances in these terms are expressed in the normalised coordinate system of each case. The complete objective is

$$
\mathcal{L}_{\text{total}} = \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{SDF}}\mathcal{L}_{\text{SDF}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}}. \tag{3.17}
$$

The surface and dense SDF terms fit the two fields to their geometric targets. The Eikonal term encourages both field gradients to have unit magnitude (Gropp et al., 2020), while the exponential off-surface term suppresses additional zero crossings. The wall term directly supervises δ at endocardial surface points. The Eikonal term is computed in full precision to keep its gradient calculation numerically stable.

Table 3.3. Purpose of the five training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathcal{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathcal{L}_{\text{SDF}}$	Fits cached query-point distances	Provides dense volumetric supervision
$\mathcal{L}_{\text{eik}}$	Encourages $\|\nabla f\| = 1$	Regularises the decoder as a signed-distance field
$\mathcal{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathcal{L}_{\text{WT}}$	Fits the predicted wall offset to wall targets	Constrains separation of the predicted fields

The retained training run uses AdamW with a learning rate of 2×10^{-5} and weight decay 5×10^{-4}. Its learning rate follows cosine annealing with warm restarts, with an initial restart period of 100 epochs. The synthetic pre-training stage lasted approximately 200 epochs; the recorded real-data fine-tuning history then continued for 776 epochs, from which the checkpoint at epoch 695 was selected by validation loss. Fine-tuning resumes from the pre-trained weights rather than starting from scratch. Training uses mixed precision and gradient clipping at 1.0. Each of the two GPUs processes a local batch of 8 cases and gradients are accumulated for two steps, giving an effective batch size of 32. A light weight-clipping strategy was used instead of the more expensive spectral normalisation, since the decoder is evaluated at many points per step and GPU memory is the main constraint.

Table 3.4. Training configuration used for the model.

Setting	Value
Training strategy	SSM pre-training, then fine-tuning on real ED/ES
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic ED and real ED/ES; targets unchanged
Optimiser	AdamW
Learning rate	2×10^{-5}
Scheduler	Cosine annealing with warm restarts ($T_0 = 100$ epochs)
Weight decay	5×10^{-4}
Training duration	Approximately 200 pre-training epochs, then 776 fine-tuning epochs (best checkpoint: 695)
Batch size	8 per GPU $\times$ 2 GPUs, accumulated for 2 steps (effective batch size 32)
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 GPUs in parallel
Inference grid	96^3

3.4.2. Inference and decoder-offset extraction

At inference time, the model turns the input contours into the latent vector z and then evaluates the decoder on a regular 96^3 grid of points. Marching cubes reads the zero-level sets of f_{endo} and f_{epi} from this grid and produces the raw endocardial and epicardial meshes (Lorensen & Cline, 1987). For evaluation, degenerate faces are removed, the largest connected component is retained, holes are filled, and a mesh is remade from a cleaned occupancy mask when needed. This post-processing makes the evaluation geometry usable for surface and thickness measurements; it is not a guarantee provided by the neural architecture, and the reported metrics do not describe an entirely unmodified decoder output. A raw analytic offset is available at every endocardial vertex because the decoder predicts δ there. Converting this uncalibrated offset to millimetres gives

$$o_{\text{raw}}(x) = s\,\delta(x, z, v(x)), \tag{3.18}$$

where s is the scale that maps normalised coordinates back to millimetres. This raw analytic offset is not measured after the fact by comparing two meshes, and it is not necessarily a Euclidean transmural separation. For visualisation, it is separately calibrated per phase by $o_{\text{cal}} = \alpha o_{\text{raw}}$, where α is the ratio between the mean segmentation-distance-transform reference and the mean raw analytic offset. This calibration fixes the mean magnitude by construction and is not an independent test of anatomical accuracy. Wall thickness is instead estimated on the reconstructed meshes by the methods in Section 3.5; the method judged most consistent in the comparison provides the final regional measurements.

The reconstruction is also compared with two non-neural reconstructions built from exactly the same SAX contour rings. The first is a radial-basis-function implicit fit. Each ring supplies on-surface constraints of value zero, and two further sets of constraints are placed 2.5 mm inside and outside along the in-plane ring normals to give the field its sign. A thin-plate-spline interpolant with a linear polynomial tail and light regularisation is fitted to those constraints and evaluated on a 1.0 mm isotropic grid spanning the observed rings, and its zero level set is extracted with marching cubes. The second is a fit of the public UK Digital Heart Project left-ventricular shape model. The mean shape is first aligned to the rings by long axis, scale and yaw, where the long axis is taken from the line through the per-slice ring centroids; correspondence, similarity registration and regularised estimation of the first 25 mode coefficients are then alternated, with the coefficients constrained to three standard deviations. Because the published model provides end-diastolic modes only, this comparator exists at end-diastole alone.

Both reconstructions pass through the same watertight repair and smoothing procedure used for the model output. Since no independent three-dimensional geometry is available, none of the three reconstructions is treated as a reference. They are compared in two ways: by the distance from the observed contour rings to each reconstructed surface, and by pairwise surface and volume agreement between them. The first quantifies how far each method honours the measured data, and the second quantifies how much the reconstruction prior changes the result. Neither isolates the contribution of individual model components.

3.5. Wall-Thickness Evaluation Protocol

The wall-thickness evaluation is performed on the model-generated geometry. The segmentation volume is used to extract contour rings and to orient the AHA regional coordinate system, but the measured surfaces are the endocardial and epicardial meshes produced by the trained model. This distinction is important: the evaluation tests thickness estimation on the reconstructed implicit model, not directly on raw segmentation meshes. Myocardial wall thickness is a clinically relevant morphological measurement, but the present evaluation does not assess diagnostic performance for infarction, scarring, or hypertrophy. The evaluation is therefore organised in three steps: first the algorithms that measure thickness on the geometry, then the resulting local three-dimensional thickness field, and finally its aggregation into the clinical AHA-17 regional summary.

3.5.1. Selected thickness methods

Four wall-thickness algorithms are evaluated on the reconstructed geometry. They are chosen to span the main methodological families used to measure thickness: volumetric distance fields, ray intersections, and partial-differential-equation (PDE) formulations. The four are the Laplace field method, the Yezzi–Prince Eulerian PDE method, the signed-distance cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. No method is assumed in advance to be the best one. All four are computed on the same geometry and compared under the same conditions, and the one that proves most consistent is then used for the regional analysis reported in Chapter 4. Each method returns local thickness values in millimetres at endocardial vertices or projected endocardial locations. The outputs are summarised with mean, median, standard deviation, fifth percentile, ninety-fifth percentile, finite-value fraction, and runtime, and are visualised as 3D colour maps and AHA-17 bullseye plots.

Table 3.5. Wall-thickness methods selected for evaluation on the model-generated LV geometry, with the Euclidean Distance Transform boundary sum kept only as a baseline.

Method	Category	Main measurement principle
Laplace field	PDE	Transmural field lines from a harmonic potential
Yezzi–Prince	PDE	Eulerian transport equations along the transmural field
SDF cone rays	Ray / SDF	Median of multiple cone rays around the surface normal
EDT boundary sum	Volumetric	Sum of endocardial and epicardial distance transforms

The EDT boundary sum computes, at each myocardial voxel x, the sum of the distances to the endocardial and epicardial surfaces,

$$t^{\mathrm{EDT}}(x) = D_{\mathrm{endo}}(x) + D_{\mathrm{epi}}(x), \tag{3.19}$$

where D_{endo} and D_{epi} are the Euclidean distance transforms of the two boundaries. This is exact for planar parallel walls but is limited by voxel resolution and slightly overestimates thickness in strongly curved regions.

The Laplace field method defines a smooth scalar field ψ over the myocardium by solving

$$\nabla^2\psi = 0, \qquad \psi|_{\text{endo}} = 0, \qquad \psi|_{\text{epi}} = 1. \tag{3.20}$$

The local wall thickness is then estimated from the gradient magnitude along the transmural field. This family of methods is attractive because harmonic field lines do not cross, but numerical errors can appear where the wall is thin or where the field gradient is small (Jones et al., 2000). The Yezzi–Prince approach addresses this by solving Eulerian transport equations along the same transmural field and combining the endocardial and epicardial travel times (Yezzi & Prince, 2003). In simplified form,

$$\nabla\psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.21}$$

where u and v are the propagation distances from opposite boundaries.

The cone-ray method uses the endocardial surface normal but reduces sensitivity to a single noisy ray. Inspired by the shape diameter function (Shapira et al., 2008), for each endocardial point several rays are cast inside a cone around the normal direction, and the median valid hit distance is used:

$$t_i^{\text{cone}} = \text{median}_{k=1}^{K}\, d_{ik}, \qquad K = 7. \tag{3.22}$$

In the notebook implementation the cone half-angle is $30°$. This makes the estimate more robust than a single normal ray while still preserving an approximately transmural interpretation.

Figure 3.8 contrasts how the four estimators measure thickness on the same wall cross-section. The EDT method sums the distances from an interior point to both boundaries; the Laplace method follows the transmural streamlines of a harmonic potential whose nested iso-surfaces are shown as dashed lines; the Yezzi–Prince method integrates two travel times u and v along that same field and adds them; and the cone-ray method casts a fan of rays around the endocardial normal and keeps the median hit distance.

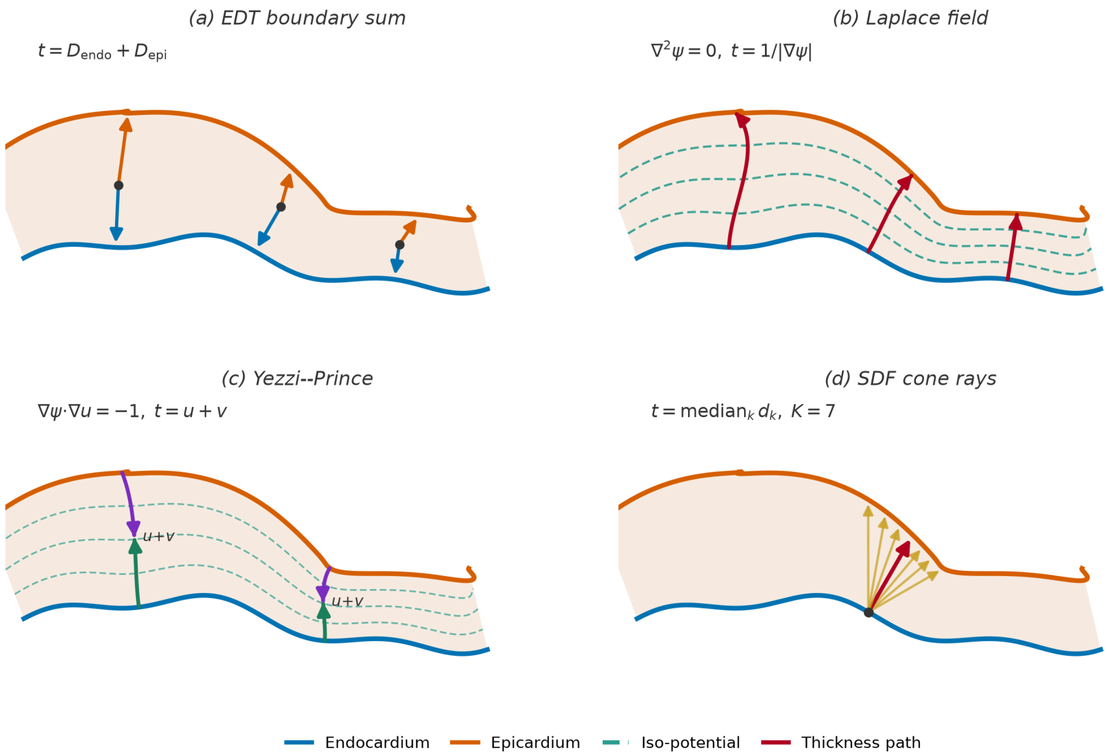

Figure 3.8. Conceptual comparison of the four wall-thickness estimators on a single myocardial cross-section, with the endocardium in blue and the epicardium in orange. (a) The EDT boundary sum adds the distances from an interior point to each boundary. (b) The Laplace field method follows the curved transmural streamlines of a harmonic potential, which cross the iso-potential surfaces (dashed lines) at right angles, and estimates thickness from the gradient magnitude. (c) The Yezzi–Prince method propagates two travel times u and v from opposite boundaries along that same curved field and sums them. (d) The SDF cone-ray method casts a fan of rays within a cone around the endocardial normal and takes the median intersection distance (highlighted), making it robust to any single ill-oriented ray.

3.5.2. Local three-dimensional wall thickness

Each of the selected methods produces, before any regional summary, a continuous thickness field defined over the whole reconstructed geometry. After the evaluation post-processing described in Section 3.4.2, the endocardial and epicardial surfaces are densely sampled and a thickness value is attached to each endocardial vertex. This produces a local three-dimensional wall-thickness map rather than a handful of per-slice numbers, as shown in Figure 3.9, where the LV surface is coloured by local thickness from the thin apical region to the thicker basal wall.

This local map is the primary output of the wall-thickness stage. Presenting the measurement first as a full surface field, and only afterwards as regional segment averages, makes the spatial distribution of thickness explicit: localised thinning or thickening remains visible on the surface even when it would be averaged away inside a single AHA segment. The colour map uses a fixed thickness range across cases so that maps from different patients and cardiac phases can be compared directly.

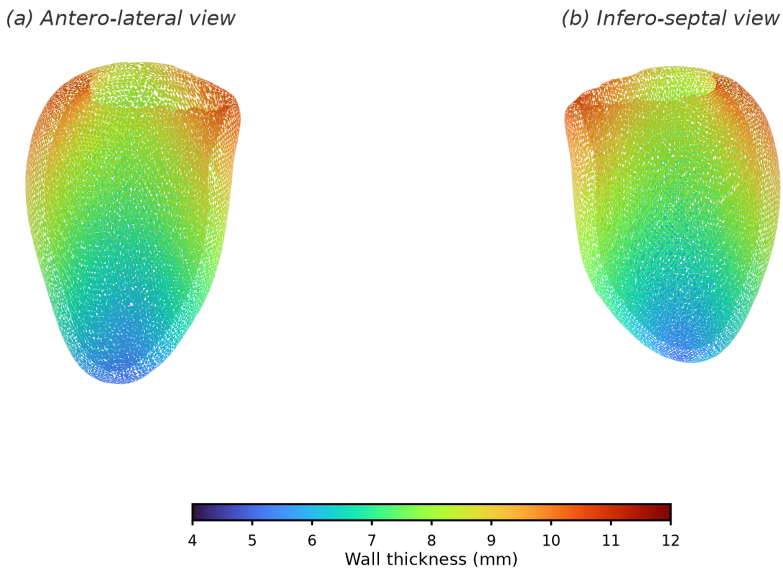

Figure 3.9. Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness on a fixed millimetre scale, so that the continuous spatial variation from the thin apex to the thicker base is visible directly on the geometry rather than only as regional averages.

3.5.3. Regional AHA-17 mapping

For the final regional analysis, the surface field from the method judged most consistent in the comparison is aggregated into the AHA 17-segment model, the accepted clinical language for reporting regional myocardial measurements (Cerqueira et al., 2002). Conventionally the wall thickness in each segment is read semi-manually from a few SAX slices, which is observer-dependent, coarsely sampled, and sensitive to the choice of slice and measurement point. Here the same segment values are produced objectively from the selected method's field of Section 3.5.2, by cutting the reconstructed surface into the AHA segments and averaging within each one.

The cut is defined in the LV's own anatomical frame, which first requires identifying the long-axis direction, distinguishing base from apex, and fixing a septal reference direction; the wall-thickness notebook includes an anatomical QA step that visualises this base/apex orientation and septal alignment before the segments are formed. The surface is then partitioned in two stages, as illustrated in Figure 3.10. First, along the long axis, the endocardial vertices are split by their normalised apex–base position into a basal ring, a mid-cavity ring, an apical ring, and a small apical cap. Second, within each ring the vertices are divided angularly around the long-axis centreline: the basal and mid rings are cut into six sectors each and the apical ring into four, while the apical cap forms the single seventeenth segment. Numbering the sectors from the anterior–septal reference and unrolling the rings onto concentric circles yields the familiar bullseye, in which the basal ring is the outer annulus, the mid ring the middle annulus, the apical ring the inner annulus, and the apical cap the central disc.

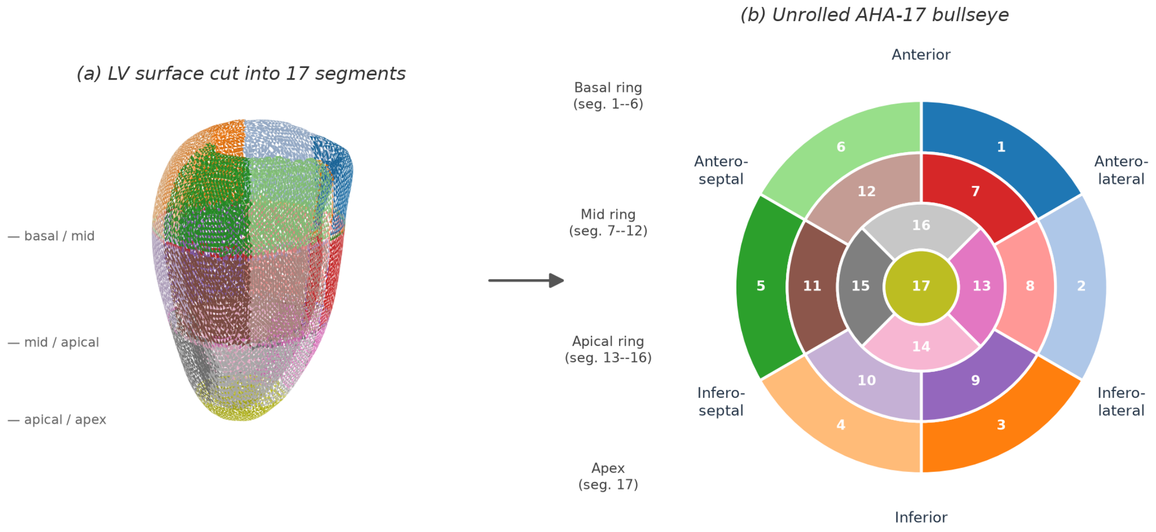

Figure 3.10. AHA-17 partitioning of the reconstructed LV surface and corresponding bullseye representation.

The AHA aggregation does not replace the per-vertex thickness maps; it provides a clinically familiar regional summary of the same local measurements. For each method, the thickness values of the vertices falling in a segment are combined using robust summaries such as the mean and percentiles, so that local outliers do not dominate the regional interpretation.

CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

The evaluation uses the final model, obtained by pre-training on the synthetic SSM meshes and then fine-tuning on the real end-diastolic (ED) and end-systolic (ES) data, as described in Chapter 3. The datasets and patient-level splits are reported in Section 3.2.3. The general reconstruction and wall-thickness evaluations use the held-out test patients from both ACDC and M&Ms-2. The pathology-stratified regional analysis uses a separate ACDC subset of 30 test patients, comprising 20 normal (NOR) and 10 hypertrophic cardiomyopathy (HCM) cases. The HCM group is included so that the model can be examined on abnormal as well as normal LV geometry. No independent 3D ground truth is available, so the following results characterise contour fidelity and inter-method consistency rather than reconstruction accuracy. Three questions guide the evaluation: how faithfully the model follows the observed LV contours and how far it agrees with alternative reconstructions of them, whether it represents the ED–ES change while maintaining a valid wall, and how closely the resulting thickness measurements agree with those obtained from an independent estimator on the same input.

4.1.1. Metrics

Reconstruction geometry is characterised with complementary surface and volume quantities, following the metric taxonomy of Taha and Hanbury (2015) and the conventions used for cardiac segmentation (Bernard et al., 2018) and for implicit-surface learning (Mescheder et al., 2019). These quantities are used in two roles: between a reconstruction and the contour rings that produced it, and between two reconstructions of the same rings. In the first role they measure fidelity to the observations; in the second they measure disagreement between methods, and are symmetric, so they belong to neither method as an error.

The *Chamfer distance* is the average distance between two surfaces, taken in both directions. It is reported for the endocardium and the epicardium in millimetres. The *contour distance* is the corresponding one-sided quantity from each observed ring point to the reconstructed surface, summarised by its mean and its 95th percentile.

The *Dice similarity coefficient* (DSC) measures the solid rather than the surface. After voxelisation it is twice the shared volume divided by the sum of the two volumes. It is computed for the endocardial cavity and for the myocardium and lies between zero and one. Enclosed *volumes* are reported in millilitres for each method separately. A volume ratio is deliberately not reported: it would require designating one reconstruction as correct, which the available data do not support.

The *watertight rate* is the fraction of reconstructions that close into a surface with no holes. It is stated in the text as a precondition rather than as a score, because a mesh that is not closed has no volume for the DSC to use.

The 95th-percentile Hausdorff distance (HD95) describes the tail of the distribution and is reported with the Chamfer distance. The average symmetric surface distance is omitted because it tracked the Chamfer distance to within 0.06 mm throughout. The intersection over union is omitted because it is a direct transformation of the DSC; the F-score depends on one selected tolerance, and normal consistency measures orientation rather than positional agreement.

Wall thickness is summarised by its mean, its standard deviation, and its 5th and 95th percentiles across the myocardial surface, all in millimetres. The percentiles describe the thin apical and thick basal extremes without being distorted by isolated outliers. When the two cardiac phases are compared, the absolute thickening $\Delta_{\mathrm{ES-ED}}$ and the relative thickening as a percentage of the end-diastolic value are also reported. These are the standard descriptors of regional contractile function.

The four wall-thickness methods are compared with one another on the reconstructed geometry, using the agreement framework of Bland and Altman (1986) instead of a comparison of mean values. The reported quantities are the *Pearson correlation coefficient* r, the *mean absolute error* (MAE) and *root-mean-square error* (RMSE) in millimetres, and the Bland–Altman *bias* with its *95% limits of agreement* (LoA) (bias $\pm 1.96\,\sigma$). Absolute consistency is summarised by the two-way *intraclass correlation coefficient* (ICC), read against the thresholds of Koo and Li (2016). These are reported together because correlation alone does not show agreement: two methods can rise and fall together and still differ by a constant amount, which only the bias reveals.

4.1.2. Comparators

The reconstruction is compared with two alternative reconstructions of the same input contour rings: a radial-basis-function implicit fit, which interpolates the rings through a smooth field, and a fit of the public statistical shape model (SSM), which regularises them towards a population prior. The two were chosen because they impose different and well-understood priors, so the spread between the three results indicates how much the reconstruction prior itself determines the geometry. Both consume the same rings as the model and pass through the same mesh repair and metric pipeline, as described in Section 3.4.2, so their numbers are directly comparable. They are evaluated only at end-diastole: the public SSM is distributed with end-diastolic modes only.

The RBF fit is a reconstruction comparator rather than ground truth. It belongs to the same family as the surrogate targets used for fine-tuning, so close agreement with the model is partly expected and is not independent corroboration. The SSM fit is independent of the training targets, but it carries a population prior that the cohort partly falls outside. Wall thickness is compared between the model, the RBF fit, and the SSM fit with the four algorithms selected in Section 3.5.1: the Laplace field method, the Yezzi–Prince method, the SDF cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. Because the SAX stacks are sparse relative to their in-plane resolution, all three reconstruction methods must infer the through-plane geometry, and their absolute millimetre values should be read with care.

4.2. Training

The final model is produced in two stages. A first stage pre-trains the signed-distance decoder for roughly 200 epochs on the synthetic SSM meshes. This stage provides a geometric prior, and its checkpoint initializes the second stage. The reported model is then fine-tuned on the real end-diastolic and end-systolic data described in Chapter 3. The available training history records the fine-tuning stage; no numerical claim is made about the pre-training loss because its per-epoch log was not retained.

Fine-tuning runs for 776 epochs with the AdamW optimiser (learning rate 2×10^{-5}, weight decay 5×10^{-4}) under a cosine-annealing schedule with warm restarts, mixed-precision arithmetic, and an effective batch size of 32. The composite objective is the five-term loss of Equation (3.17), combining surface, dense SDF, Eikonal, off-surface, and wall-thickness supervision. Figure 4.1 plots the total loss and its principal components together with two optimisation diagnostics. The total training loss decreases from 10.07 at the first fine-tuning epoch to 0.91, and the validation loss settles at 0.70, with the best checkpoint selected at epoch 695 before mild over-fitting appears. The surface and Eikonal terms fall by more than an order of magnitude, indicating successful optimisation of the surface-fitting and Eikonal objectives. Consistently, the mean gradient norm $\|\nabla f\|$ stabilises at 0.996, close to the unit value encouraged by the Eikonal term.

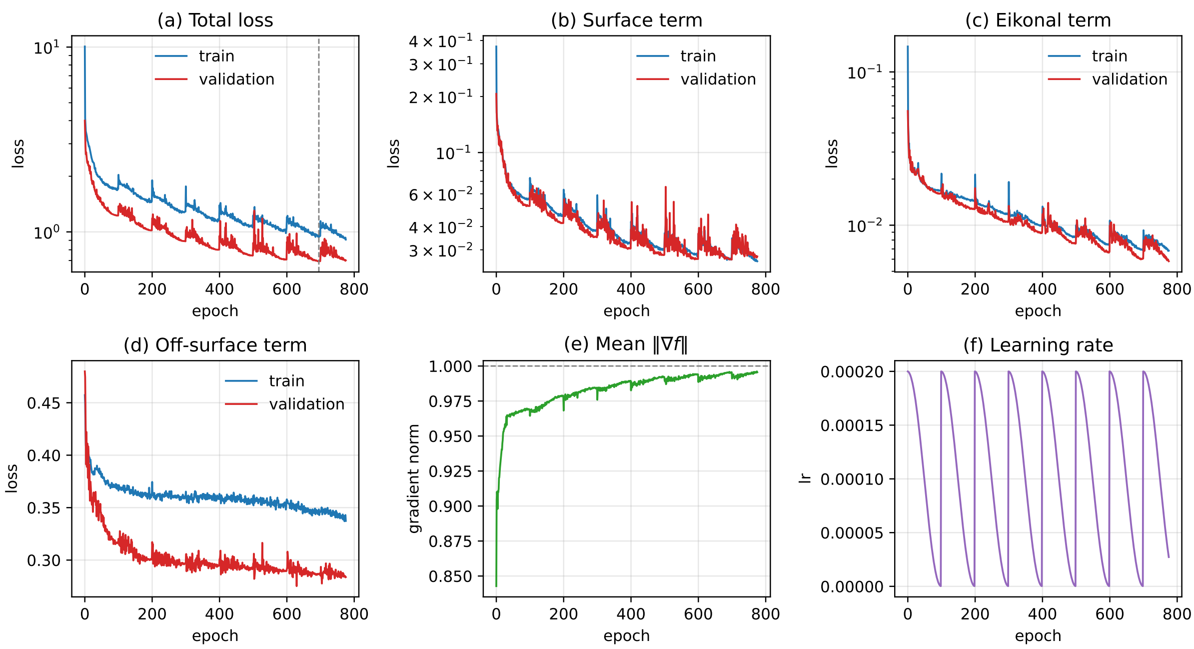

Figure 4.1. Training history. Panels (a)–(d) show the total loss and its surface, Eikonal, and off-surface components for the training and validation splits; the dashed vertical line marks the best epoch. Panels (e)–(f) show the mean field gradient norm, whose target is one, and the cosine-annealing learning-rate schedule. The synthetic pre-training stage is not shown because its optimisation history was not retained.

4.3. Results

This section reports first the geometric quality of the reconstruction, then the wall-thickness measurements obtained from the reconstructed geometry, and finally the regional analysis over the AHA 17-segment model.

4.3.1. Reconstruction Consistency

The following results characterise contour fidelity and inter-method consistency rather than reconstruction accuracy. Contour fidelity measures how closely each surface follows its input rings, while pairwise agreement measures how strongly the reconstruction prior changes the completed geometry. All values below are end-diastolic and cover the held-out ACDC and M&Ms-2 test cohort, and every surface of every method is watertight after the shared repair stage.

4.3.1.1. *Reconstruction Methods* This comparison covers how each method follows the observed contours, the volumes it encloses, and how its complete reconstruction agrees with the other two methods.

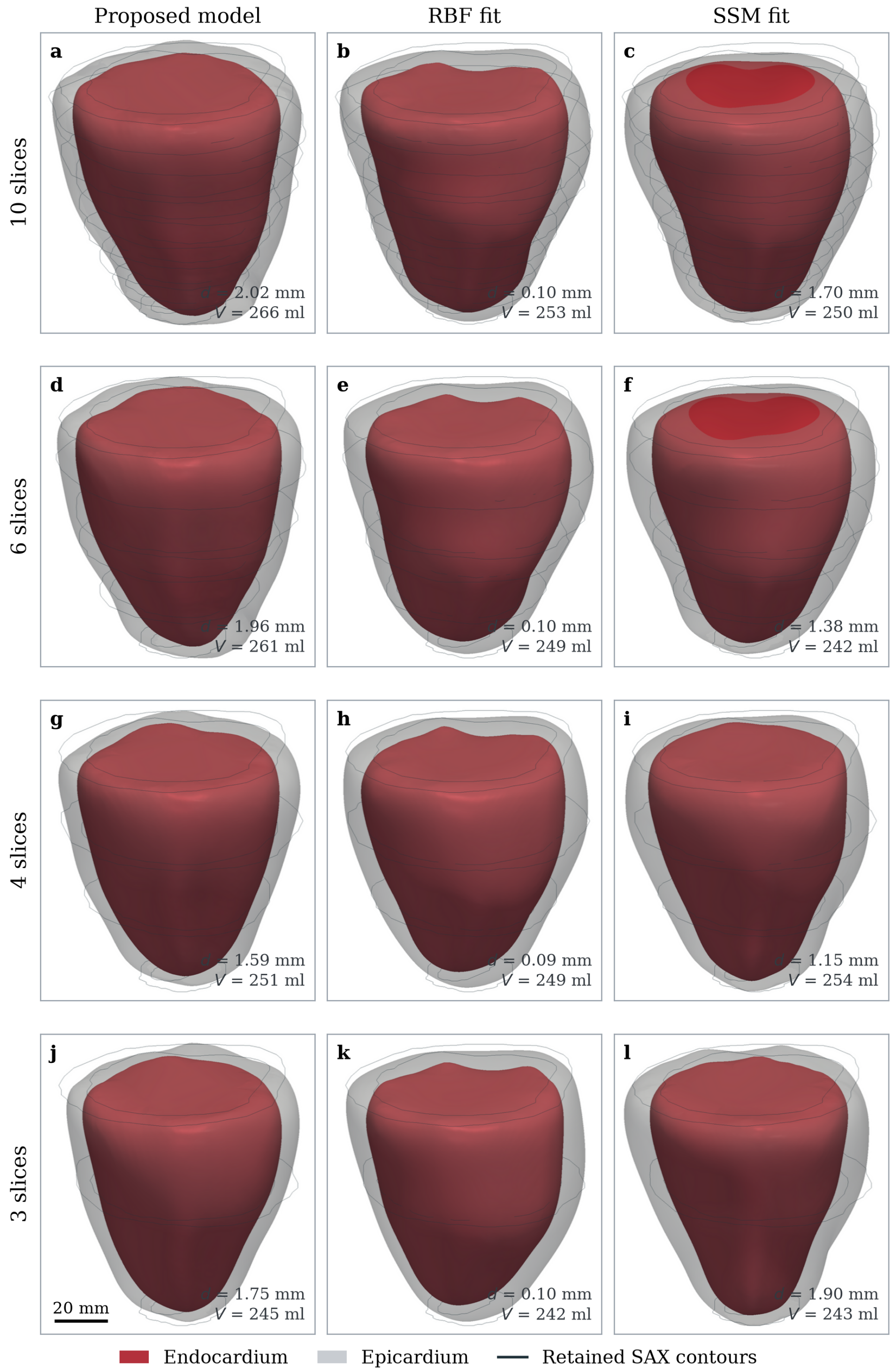

Figure 4.2. Left-ventricular surfaces recovered by the three methods from the same end-diastolic SAX stack after it is progressively decimated. Columns are the proposed model, the RBF implicit fit and the SSM fit; rows are the number of retained slices, from the complete ten-slice stack down to three. The most basal and the most apical slice are always kept, so every panel spans the same axial range. The opaque red surface is the endocardium, the translucent grey shell the epicardium, and the thin dark rings the retained input contours. All twelve panels share one camera, a parallel projection and one scale, with the apex downwards. Each panel reports the mean endocardial contour distance d and the enclosed endocardial volume V, measured on the untrimmed and unsmoothed mesh. The case shown is an illustrative ACDC patient with dilated cardiomyopathy; this is a single-case demonstration and not a cohort result.

Figure 4.2 first supports a direct comparison between methods through its complete ten-slice row. All three recover the same overall ventricular shape from identical observations, but they encode different priors. The RBF fit interpolates the retained contours, the SSM fit regularises the surface towards a population shape, and the proposed model combines a learned prior with conditioning on the observed rings. The cohort-level differences between these reconstructions are examined first through contour fidelity and pairwise geometric agreement. The remaining rows of the figure are then analysed separately as a sensitivity test of inference under reduced slice density.

The first part of the comparison considers fidelity to the observed contours and the enclosed volumes. Table 4.1 reports the distance from each input contour ring to the surface reconstructed from it, together with the enclosed volumes. The implicit fit reproduces the rings almost exactly, at 0.13 and 0.11 mm on the two surfaces. That figure must not be read as an order-of-magnitude advantage. The fit is an interpolant whose defining constraint is that the field vanish at exactly those contour points, so reproducing them is a boundary condition of the method and the measured residual is the solver's, not the reconstruction's. Any interpolating method scores near zero here and any regularising method does not, whatever their relative merits between the slices. The quantity is reported because it verifies that each method honours its input and because it separates interpolating from regularising behaviour, not because it ranks the methods.

Two further limits apply. The rings are separated by several millimetres, so a metric anchored on them says nothing about the intervening volume, which is where the reconstruction problem actually lies. And exact fidelity to the rings is not unambiguously desirable, because the contours carry annotation noise and in-plane discretisation: a surface that follows them to 0.13 mm follows their errors as well.

The model deviates from the rings by 1.50 mm on the endocardium and 1.96 mm on the epicardium, which is of the order of the in-plane pixel size of 1.37 to 1.88 mm, and is therefore consistent with a surface that smooths annotation noise rather than one that misses the observations. The SSM fit deviates by a similar amount on average, at 1.82 and 1.91 mm, but far less predictably: its per-patient endocardial figure spans 0.92 to 3.44 mm against 1.20 to 2.14 mm for the model.

Within the 30-patient ACDC pathology subset, the diagnostic split explains that spread. The SSM fit is close to the observations on the normal cases, at 1.42 mm, and drifts on the hypertrophic ones, at 2.63 mm, which is the expected behaviour of a population prior asked to represent anatomy outside its training variation. The model is comparatively stable across the two groups, at 1.45 and 1.60 mm, and the implicit fit is insensitive to pathology by construction.

Enclosed volumes tell a complementary story. The three methods agree closely on the cavity, at 119, 119 and 127 ml, but diverge on the myocardium, at 136, 115 and 88 ml. The wall, not the cavity, is where the reconstruction priors disagree, which matters directly for the thickness measurements reported in Section 4.3.2.

Table 4.1. Distance from the input SAX contour rings to the surface reconstructed from them, and the enclosed volumes, at end-diastole over the held-out ACDC and M&Ms-2 test cohort. Values are mean $\pm$ standard deviation of the per-patient statistic; p_{95} is the cohort mean of the per-patient 95th percentile. The RBF fit interpolates the contour points by construction.

Method	Surface	Contour distance (mm)	p_{95} (mm)	Volume (ml)
Proposed model	Endocardium	1.50 ± 0.23	3.56	119.3 ± 31.3
	Epicardium	1.96 ± 0.28	5.23	255.3 ± 76.7
RBF fit	Endocardium	0.13 ± 0.01	0.30	118.6 ± 30.2
	Epicardium	0.11 ± 0.01	0.26	233.5 ± 71.4
SSM fit	Endocardium	1.82 ± 0.67	5.51	126.9 ± 32.2
	Epicardium	1.91 ± 0.76	6.23	215.2 ± 56.3

The second part considers agreement between the complete reconstructed surfaces and volumes. Table 4.2 compares the three reconstructions with one another. Because none of them is a reference, each entry is a symmetric disagreement between two methods rather than an error of either. The average symmetric surface distance is omitted because it differed from the Chamfer distance by less than 0.06 mm for every pair.

The model and the implicit fit are the closest pair, differing by 1.57 mm on the endocardium and 1.90 mm on the epicardium, with a myocardial Dice of 0.77. The two pairs involving the SSM are further apart, at 2.08 and 2.01 mm on the endocardium. The ordering is the same for every surface metric, which places the learned reconstruction nearer to a smooth interpolant of the observations than to a population prior.

Two features of these numbers deserve emphasis. The disagreement between any two methods, between 1.6 and 2.1 mm on the endocardium, is comparable to the in-plane pixel size and far smaller than the 10 mm slice spacing, so all three priors agree about the ventricle to within roughly one pixel where the data constrain them. Cavity agreement is uniformly high, at Dice 0.88 to 0.91, while myocardial agreement is markedly lower, at 0.69 to 0.77, and its patient-level range is wide: myocardial Dice falls to 0.58 for the worst SSM comparison. The reconstructions therefore concur about the cavity and disagree about the wall, which is the quantity this thesis ultimately measures. That spread is a floor on the uncertainty of any wall-thickness value reported later in this chapter, and it cannot be reduced by choosing one of these methods as though it were correct.

Table 4.2. Pairwise agreement between the three reconstructions of the same end-diastolic contour rings, over the held-out ACDC and M&Ms-2 test cohort. Values are mean $\pm$ standard deviation of the per-patient statistic. Every entry is a symmetric disagreement between two methods, not an error, because no method is a reference. Distances are in millimetres; Dice is dimensionless.

Pair	Metric	Endocardium/cavity	Epicardium/myocardium
Model vs. RBF fit	Chamfer (mm)	1.57 ± 0.18	1.90 ± 0.24
	HD95 (mm)	4.24 ± 0.86	6.07 ± 0.94
	Dice	0.91 ± 0.02	0.77 ± 0.05
Model vs. SSM fit	Chamfer (mm)	2.08 ± 0.51	2.59 ± 0.73
	HD95 (mm)	5.88 ± 1.91	6.72 ± 2.32
	Dice	0.88 ± 0.03	0.69 ± 0.04
RBF fit vs. SSM fit	Chamfer (mm)	2.01 ± 0.67	2.05 ± 0.64
	HD95 (mm)	5.88 ± 2.95	6.62 ± 2.67
	Dice	0.89 ± 0.04	0.76 ± 0.06

4.3.1.2. *Slice Sensitivity* Having established the differences between reconstruction methods on the complete stacks, the remaining rows of Figure 4.2 examine how their inference changes when through-plane information is removed. The ten-slice end-diastolic stack is reduced to six, four and three slices while always retaining the most basal and most apical slice. The axial support is therefore fixed, while the largest through-plane gap increases from 10 to 45 mm. All three methods continue to produce closed, plausible surfaces with only three slices, so none fails outright under this reduction.

Table 4.3 quantifies the change by comparing each decimated reconstruction with the surface produced by the same method from all ten slices. These within-method deviations measure self-consistency rather than anatomical error and isolate sensitivity to missing slices from the between-method differences reported above.

Table 4.3. Deviation from each method's complete ten-slice reconstruction for one illustrative end-diastolic ACDC case. Distances are in millimetres and Dice is dimensionless; ΔV_{myo} is the change in myocardial volume relative to the ten-slice reconstruction. All meshes are watertight, and metrics use the untrimmed and unsmoothed surfaces.

		Chamfer (mm)		Dice		
Method	**Slices**	**Endo.**	**Epi.**	**Cavity**	**Myoc.**	ΔV_{myo} **(%)**
Proposed model	6	0.84	0.92	0.972	0.892	-7.2
	4	1.45	1.76	0.945	0.772	-16.3
	3	2.01	2.58	0.921	0.655	-22.7
RBF fit	6	0.83	0.83	0.968	0.887	0.0
	4	1.15	0.88	0.959	0.858	-1.8
	3	1.45	1.32	0.947	0.805	-6.2
SSM fit	6	0.89	0.95	0.968	0.883	-1.8
	4	1.46	1.39	0.942	0.812	-3.8
	3	1.94	1.92	0.921	0.775	$+2.6$

Surface displacement increases progressively as slices are removed. At three slices, endocardial Chamfer distance relative to the complete-stack result is 2.01 mm for the model, 1.45 mm for the RBF fit and 1.94 mm for the SSM fit. Cavity Dice remains at least 0.921, but myocardial Dice falls to 0.655, 0.805 and 0.775, respectively. The cavity is therefore more stable than the wall under slice decimation, consistent with the cohort comparison in Table 4.2.

The model is the most slice-dependent in this case: its myocardial volume falls by 22.7% from ten to three slices, compared with 6.2% for the RBF fit, while the SSM fit changes by $+2.6$%. The apparent stability of the SSM fit does not establish greater anatomical accuracy, because its population prior can remain stable by responding less strongly to the retained contours. The experiment instead shows that slice density is a material source of uncertainty for absolute wall geometry, especially for the proposed model.

4.3.2. Wall-Thickness Measurement

The four selected methods are applied to the proposed reconstruction, the RBF fit, and the SSM fit of every patient at end-diastole. Table 4.4 reports, for each estimator and reconstruction, the cohort average of the per-patient mean, standard deviation, and 5th and 95th percentiles.

The three non-ray methods agree on the magnitude of the wall within each reconstruction. Their means range from 9.18 to 10.01 mm on the proposed reconstruction, from 8.36 to 8.86 mm on the RBF fit, and from 6.48 to 7.10 mm on the SSM fit. The reconstruction prior therefore changes the absolute thickness scale even when the estimator is held fixed. The SDF cone-ray method gives the largest value on the proposed reconstruction and the RBF fit, at 12.47 and 9.38 mm, respectively, but not on the SSM fit.

Dispersion separates the methods the same way. The per-patient standard deviation is 1.64 mm for Yezzi–Prince, 1.66 mm for the EDT boundary sum and 2.14 mm for the Laplace field on the proposed reconstruction, but 4.87 mm for the cone-ray estimator. Its 5th-to-95th-percentile span, 8.01 to 20.68 mm, is about twice as wide as that of any other estimator on that geometry. The ray-based estimator is therefore both the highest and the noisiest of the four on the model output.

Table 4.4. Wall-thickness statistics at end-diastole over the held-out ACDC and M&Ms-2 test cohort. Each entry is the cohort average of the corresponding per-patient statistic, in millimetres, before any calibration.

Method	Geometry	Mean (mm)	SD (mm)	p_5 (mm)	p_{95} (mm)
Laplace field	Model	9.18	2.14	4.75	12.20
	RBF fit	8.36	2.21	5.03	12.29
	SSM fit	6.48	1.38	3.87	8.43
Yezzi–Prince	Model	10.01	1.64	7.41	12.74
	RBF fit	8.86	2.15	5.57	12.71
	SSM fit	7.10	1.23	4.75	8.88
SDF cone rays	Model	12.47	4.87	8.01	20.68
	RBF fit	9.38	2.49	5.63	13.80
	SSM fit	7.11	1.50	4.12	9.13
EDT boundary sum	Model	9.55	1.66	6.62	12.17
	RBF fit	8.51	2.00	5.42	12.05
	SSM fit	7.02	1.22	4.65	8.77

Applying the same estimator to all three geometries isolates the effect of the reconstruction prior. Relative to the model, the RBF fit is thinner by 0.82 mm for the Laplace field, 1.16 mm for Yezzi–Prince, 3.09 mm for SDF cone rays, and 1.03 mm for the EDT boundary sum. The SSM fit is thinner by 2.70, 2.91, 5.36, and 2.53 mm, respectively. Per-patient means remain strongly correlated between the model and RBF fit ($r = 0.99$, 0.99, 0.95, and 0.99), whereas agreement with the SSM fit is more moderate ($r = 0.72$, 0.77, 0.67, and 0.77). The RBF fit therefore largely preserves between-patient variation while shifting the absolute scale; the stronger SSM prior changes both the scale and, to a lesser extent, the patient ordering.

Table 4.5 compares the three other methods with the Laplace field point by point over the myocardial surface. The two remaining non-ray methods track it closely. The EDT boundary sum differs from it by 0.81 mm on average with a bias of $+0.37$ mm, and Yezzi–Prince by 0.89 mm with a bias of $+0.83$ mm; both reach ICC $= 0.77$. The cone-ray method differs by 3.38 mm on average, with a bias of $+3.31$ mm, limits of agreement of ± 12.74 mm and ICC $= 0.28$. Read against the thresholds of Koo and Li (2016), the first two show moderate to good agreement with the Laplace field and the third poor agreement. The correlations make the same point in a different way: all three exceed $r = 0.44$, but r alone would not have revealed a bias of more than three millimetres.

The Laplace field is the method used for the regional analysis of Section 4.3.3. The reason is methodological rather than numerical. Although introduced for cortical thickness rather than cardiac imaging (Jones et al., 2000), it provides a transferable construction for paired tissue boundaries. It measures thickness along the streamlines of a harmonic field spanning the two surfaces, so the path stays inside the wall wherever the wall curves. A straight ray does not: it leaves the myocardium early or late in curved regions, which is what the cone-ray numbers above show. The results here are consistent with that choice, since the Laplace field sits with the two non-ray methods rather than with the outlier.

The decoder returns a continuous wall-offset field rather than a geometric thickness measurement. Figure 4.3 visualises this field for the representative case at both cardiac phases, in the same form as the methodology illustration of Figure 3.9 but computed on the model output instead of on the SSM mean mesh. The decoder offset is defined only up to the normalisation applied to the input contours, so its absolute level is calibrated per phase to the mean of the distance-transform reference measured on the input segmentation, exactly as the cohort pipeline does; the calibration factor is close to two for both phases. The figure is therefore a calibrated analytic offset map, not a wall-thickness measurement or an independently predicted absolute magnitude. The residual circumferential banding visible on the surfaces marks the through-plane spacing of the input SAX slices, where the field is interpolated between observations.

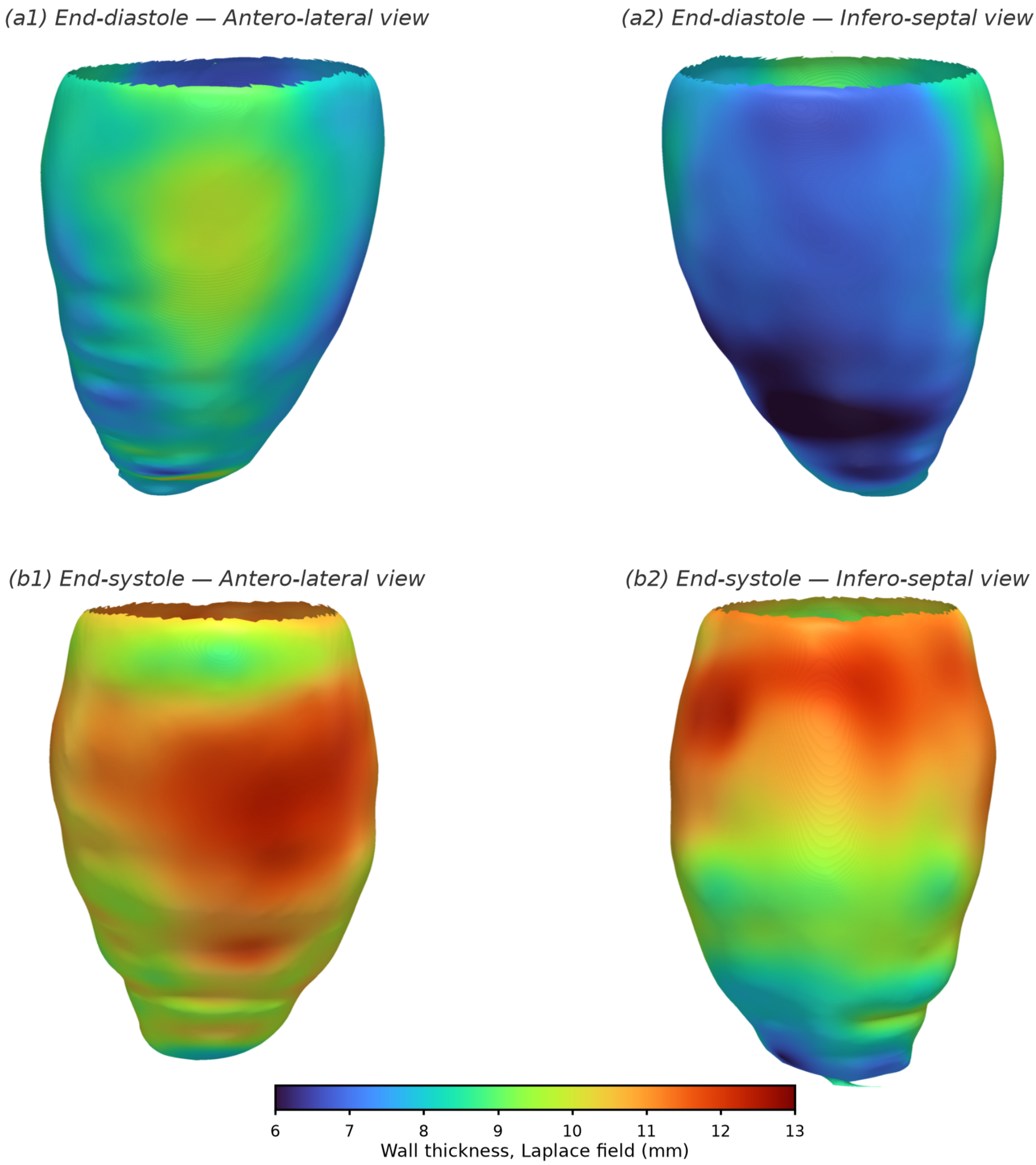

Figure 4.3. Calibrated analytic offset map on the reconstructed endocardial surface of one representative case from the ACDC cohort, at end-diastole (top) and end-systole (bottom), each seen from two opposing viewpoints. Every vertex is coloured by the decoder wall offset evaluated at that point, calibrated per phase to the distance-transform reference of the input segmentation, on a scale shared by all four panels. The absolute colour level is therefore inherited from that reference. The surface is trimmed at the most basal input contour, so the base ends in an open rim and only the observed part of the wall is shown; above that plane the field is pure extrapolation.

Figure 4.3 and Table 4.8 intentionally report different quantities. The figure visualises the calibrated decoder offset to show the spatial pattern emitted directly by the model. The table reports Laplace-derived wall thickness measured between the post-processed endocardial and epicardial meshes. The offset map is therefore a qualitative model-output visualisation; it does not provide the values used for the AHA-17 analysis.

Table 4.5. Agreement of each wall-thickness method with the Laplace field, over the myocardial surface points of the held-out ACDC and M&Ms-2 test cohort. MAE, RMSE, bias and the 95% LoA are in millimetres; the Pearson correlation r and the intraclass correlation coefficient (ICC) are dimensionless. The Laplace field is the common comparator, not a ground truth.

Method	r	**MAE (mm)**	**RMSE (mm)**	**Bias (95% LoA) (mm)**	**ICC**
Yezzi–Prince	0.80	0.89	2.14	$+0.83$ (± 3.86)	0.77
SDF cone rays	0.44	3.38	7.29	$+3.31$ (± 12.74)	0.28
EDT boundary sum	0.78	0.81	2.07	$+0.37$ (± 3.99)	0.77

Table 4.6 quantifies the patient-level agreement behind the cohort means. The RBF fit remains close to the model for the three non-ray methods, with biases of $+0.82$ to $+1.16$ mm and correlations of $r = 0.99$. The cone-ray bias is larger at $+3.10$ mm. The SSM fit differs more strongly for every estimator, with biases of $+2.53$ to $+5.36$ mm and correlations from $r = 0.67$ to 0.77. These results confirm that both the reconstruction prior and the thickness estimator affect the reported scale.

Table 4.6. Patient-level agreement between wall thickness measured on the proposed reconstruction and on the two fitting baselines at end-diastole. Bias is model minus comparator; bias and RMSE are in millimetres, and the Pearson correlation r is dimensionless.

	Model vs. RBF fit			**Model vs. SSM fit**		
Method	**Bias (mm)**	**RMSE (mm)**	r	**Bias (mm)**	**RMSE (mm)**	r
Laplace field	$+0.82$	0.86	0.99	$+2.70$	3.13	0.72
Yezzi–Prince	$+1.16$	1.23	0.99	$+2.91$	3.49	0.77
SDF cone rays	$+3.10$	3.81	0.95	$+5.36$	6.64	0.67
EDT boundary sum	$+1.03$	1.10	0.99	$+2.53$	3.04	0.77

The fitting-baseline comparison is limited to ED because the public SSM provides no ES modes. Phase-dependent wall thickening is therefore reported on the reconstructed geometry in the regional analysis below.

4.3.3. Regional Analysis over the AHA 17-Segment Model

To localise the measurements anatomically, the reconstructed surface is partitioned into the seventeen segments of the American Heart Association (AHA) model, assigning each myocardial vertex to a basal, mid-cavity, apical, or apical-cap segment from its normalised long-axis height and its circumferential angle relative to the right-ventricular insertion.

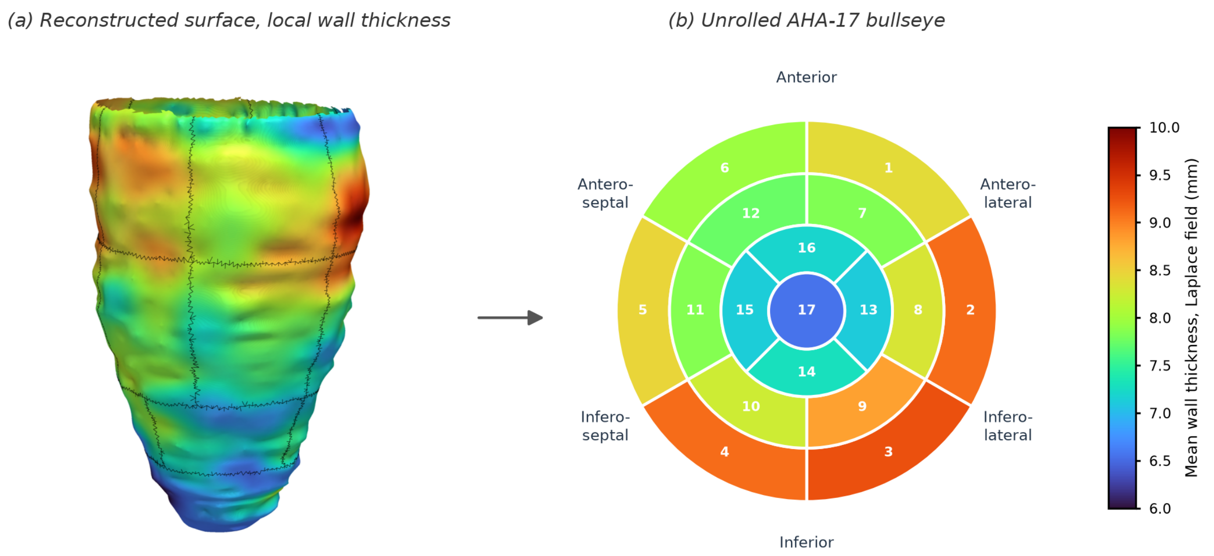

Figure 4.4. AHA 17-segment analysis of the reconstructed left-ventricular geometry at end-diastole for the representative case. Panel (a) shows the reconstructed surface partitioned into the seventeen segments, each coloured by its mean wall thickness measured with the Laplace field method and separated from its neighbours by a black boundary line; panel (b) unrolls the same segment values onto the standard bullseye, where basal segments form the outer ring (1–6), mid-cavity segments the middle ring (7–12), apical segments the inner ring (13–16), and the apical cap the centre (17). The two panels share the colour scale, so the bullseye is a flattening of the geometry beside it rather than an independent summary.

Figure 4.4 shows the partition applied to the reconstructed geometry of the representative case together with a bullseye of the Laplace-derived wall thickness. The regional results are then read in two steps.

Table 4.7 first reports the regional pattern over the 30-patient ACDC pathology subset and its change between phases. Table 4.8 then separates its normal and hypertrophic patients, showing whether the model reacts to an abnormal ventricle or simply reproduces an average one. All values are measured with the Laplace field, for the reasons given in Section 4.3.2.

Table 4.7. Regional AHA 17-segment wall thickness on the reconstructed geometry over the 30-patient ACDC pathology subset.

#	Segment	ED (mm)	ES (mm)	Δ (%)
1	Basal anterior	9.99	13.71	$+37.1$
2	Basal anteroseptal	10.15	13.40	$+32.1$
3	Basal inferoseptal	9.89	13.43	$+35.7$
4	Basal inferior	9.09	13.81	$+51.9$
5	Basal inferolateral	8.93	14.27	$+59.8$
6	Basal anterolateral	9.53	13.87	$+45.5$
7	Mid anterior	9.19	13.79	$+50.0$
8	Mid anteroseptal	10.37	14.68	$+41.6$
9	Mid inferoseptal	10.19	15.05	$+47.6$
10	Mid inferior	9.25	14.41	$+55.8$
11	Mid inferolateral	9.09	14.96	$+64.6$
12	Mid anterolateral	8.99	14.17	$+57.6$
13	Apical anterior	8.50	13.63	$+60.3$
14	Apical septal	8.77	13.86	$+58.1$
15	Apical inferior	8.44	13.29	$+57.5$
16	Apical lateral	8.43	13.78	$+63.6$
17	Apex	7.76	11.69	$+50.7$
Overall		**9.21**	**13.87**	$+$**50.6**

Note. Values are means for the 30 ACDC test patients used in the pathology analysis, measured with the Laplace field. ED and ES denote end-diastole and end-systole, respectively; Δ is the relative thickening from ED to ES.

Over the ACDC pathology subset the reconstructed wall averages 9.21 mm at ED and 13.87 mm at ES, corresponding to 50.6% relative thickening. At ED the septal mid-cavity segments are the thickest, while the apical cap is the thinnest at 7.76 mm. Every segment thickens between phases; the smallest relative change is 32.1% in the basal anteroseptal segment and the largest is 64.6% in the mid inferolateral segment. These phase values describe the model output and are not an accuracy comparison, because neither fitting baseline provides a matched regional ES evaluation.

Table 4.8. Regional AHA 17-segment wall thickness on the reconstructed geometry for the normal and hypertrophic groups.

#	Segment	Normal (NOR, $n = 20$) ED (mm)	ES (mm)	Δ (%)	Hypertrophic (HCM, $n = 10$) ED (mm)	ES (mm)	Δ (%)
1	Basal anterior	8.72	12.25	$+40.4$	12.54	16.62	$+32.6$
2	Basal anteroseptal	8.81	12.10	$+37.4$	12.83	16.02	$+24.9$
3	Basal inferoseptal	8.68	11.96	$+37.8$	12.32	16.35	$+32.8$
4	Basal inferior	8.16	12.53	$+53.6$	10.96	16.37	$+49.4$
5	Basal inferolateral	7.94	12.81	$+61.3$	10.91	17.19	$+57.5$
6	Basal anterolateral	8.38	12.12	$+44.6$	11.83	17.37	$+46.9$
7	Mid anterior	7.94	12.10	$+52.5$	11.70	17.16	$+46.6$
8	Mid anteroseptal	8.55	12.98	$+51.7$	13.99	18.10	$+29.4$
9	Mid inferoseptal	8.57	12.98	$+51.4$	13.43	19.17	$+42.8$
10	Mid inferior	7.93	12.46	$+57.1$	11.89	18.32	$+54.1$
11	Mid inferolateral	7.96	13.02	$+63.5$	11.35	18.84	$+66.0$
12	Mid anterolateral	7.92	12.39	$+56.4$	11.13	17.73	$+59.3$
13	Apical anterior	7.28	11.75	$+61.4$	10.96	17.40	$+58.7$
14	Apical septal	7.49	12.00	$+60.2$	11.33	17.59	$+55.3$
15	Apical inferior	7.28	11.72	$+61.1$	10.76	16.43	$+52.6$
16	Apical lateral	7.33	12.03	$+64.0$	10.61	17.28	$+62.9$
17	Apex	6.80	10.32	$+51.8$	9.67	14.71	$+52.1$
Overall		**7.99**	**12.21**	$+$**52.9**	**11.68**	**17.23**	$+$**47.5**

Note. Values are group means measured with the Laplace field. ED and ES denote end-diastole and end-systole, respectively; Δ is the relative thickening from ED to ES. These measurements summarise the model output reconstructed from sparse SAX contours and are not an independent anatomical reference.

The regional pattern is internally coherent in both groups. In the normal cases the end-diastolic thickness varies little around the base and the mid-cavity, from 7.92 to 8.81 mm, and tapers towards the apex, where the apical cap at 6.80 mm is the thinnest region. Every segment thickens from end-diastole to end-systole, and the thickening is graded along the long axis: the apical segments thicken by 52% to 64%, the mid-cavity segments by 51% to 64% and the basal segments by 37% to 61%. Over the normal group the mean thickness rises from 7.99 mm to 12.21 mm, a thickening of 52.9%.

The hypertrophic group is thicker in every segment. The largest differences are in the mid-cavity anteroseptal and inferoseptal segments, where the HCM group reaches 13.99 and 13.43 mm compared with 8.55 and 8.57 mm in the normal group. Table 4.9 evaluates this separation at the patient level rather than treating the 17 segments as independent observations.

Table 4.9. Patient-level contrasts between the normal and HCM groups. Values are mean $\pm$ standard deviation. Differences are HCM minus NOR with a 95% percentile-bootstrap confidence interval from 10 000 patient-level resamples. Positive Hedges' g indicates a larger value in HCM.

Endpoint	NOR	HCM	Difference (95% CI)	Hedges' g
Mean ED AHA-17 thickness (mm)	7.99 ± 0.88	11.66 ± 1.66	$+3.67$ $[2.54,\ 4.67]$	3.01
Maximum ED segment (mm)	9.07 ± 0.98	14.44 ± 1.94	$+5.37$ $[4.11,\ 6.56]$	3.82
Systolic thickening (percentage points)	53.2 ± 11.6	47.9 ± 11.5	-5.3 $[-13.9,\ 3.0]$	-0.45

The two end-diastolic confidence intervals exclude zero and their effect sizes are large, supporting descriptive separation in absolute thickness. In contrast, the interval for relative systolic thickening includes zero; this cohort does not establish a clear group difference in proportional thickening. Despite the maximum-segment contrast, none of the normal cases and only three of the ten HCM cases exceed the 15 mm clinical threshold. The measurements must therefore not be interpreted as a diagnostic result.

Figure 4.5 shows the patient-level spread behind these estimates. Absolute ED thickness separates the groups more clearly than relative systolic thickening, for which the distributions overlap.

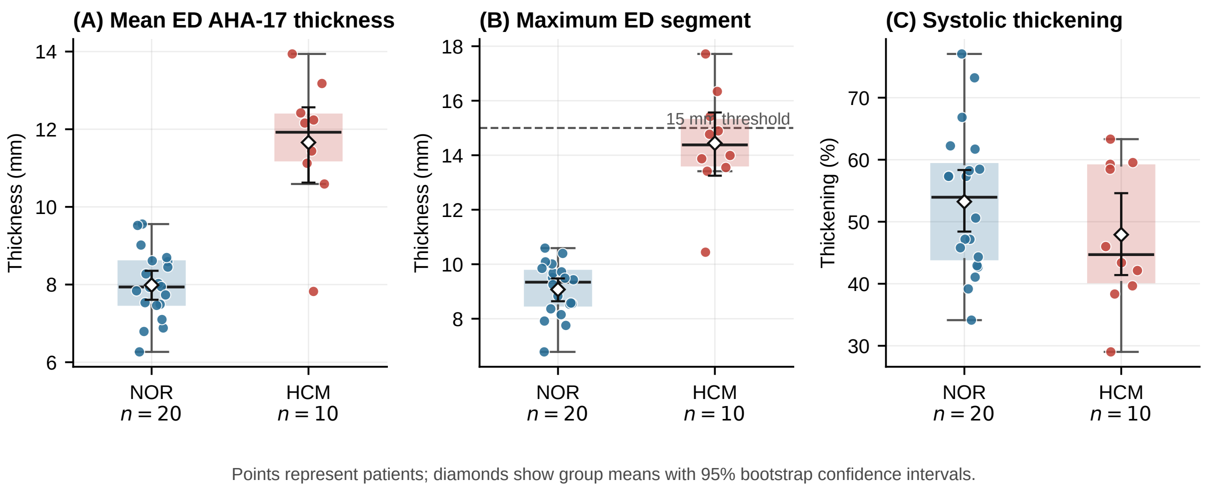

Figure 4.5. Patient-level AHA-17 wall-thickness outcomes for the normal and HCM groups. The dashed line marks the 15 mm threshold for the maximum ED segment.

One property of this analysis should be stated explicitly. Each patient contributes seventeen segment values, so the segment observations are not independent of one another. Segments from the same heart share the same reconstruction, the same input contours and the same anatomy. For this reason the segment values are used as a descriptive regional summary, and the group comparisons above are made at the patient level, where each patient counts once. The segment-level numbers should not be treated as if they came from independent observations.

4.4. Discussion

The results provide a direct answer to each research question stated in Section 1.3.

For **RQ1**, the model can reconstruct continuous, closed 3D LV surfaces from sparse SAX contours within the held-out ACDC and M&Ms-2 test cohort. It follows the observed contour rings to within about the in-plane pixel size, and every extracted endocardial and epicardial surface is watertight after repair (Table 4.1). The complete-stack row of Figure 4.2 shows that the three reconstructions recover the same overall geometry from identical contour input, and Table 4.2 puts that impression on a scale: the model sits within 1.6 mm of a smooth interpolant of the observations and within 2.1 mm of a population shape prior. Because no reference geometry exists, RQ1 is answered through contour fidelity, closure, and inter-method consistency. Closure additionally depends on the repair stage described in Chapter 3; it is not guaranteed by surface extraction alone. The subsequent slice-density analysis shows that all three methods still return closed surfaces from three slices, although the model's wall geometry changes most strongly (Table 4.3).

For **RQ2**, the model represents separate endocardial and epicardial surfaces at both ED and ES while maintaining a positive wall offset at every evaluated point. The paired surfaces do not collapse, and they reproduce the expected smaller cavity and thicker wall at ES. The answer is therefore positive at the level of geometric representation. The regional measurements increase from ED to ES in every segment (Table 4.7), but no matched ES SSM comparator is available, so phase-specific reconstruction accuracy cannot be established.

For **RQ3**, measurements on the reconstructed surfaces track those from the RBF fit across patients, although their absolute values differ. The four estimators show correlations of $r = 0.95$ to 0.99 between those two reconstructions, while correlations with the SSM fit range from $r = 0.67$ to 0.77 (Table 4.4). The reconstruction therefore preserves relative differences most strongly against the smooth interpolating fit. The absolute scale remains dependent on both the reconstruction prior and the estimator. This answers RQ3 in terms of computational agreement.

Taken together, the demonstrated result is geometric and computational: the model completes sparse contours into paired watertight surfaces that follow the observations, preserves relative between-patient thickness variation, supports regional summaries, and stays within about two millimetres of two conventional reconstructions of the same data.

4.5. Threats to Validity

The most important limitation concerns the reference. No public dataset provides independently acquired 3D ground-truth surfaces for these scans, so the reconstruction is evaluated through fidelity to the observed contour rings and agreement with alternative reconstructions of those rings. The pairwise Chamfer distances therefore measure disagreement between reconstruction priors, not distance from true anatomy. The same caveat applies to the wall-thickness comparison: the input is a stack of contours separated by several millimetres, so no method used in this chapter carries measured through-plane information, and the way that gap is inferred shifts the measured thickness. Absolute millimetre values should therefore not be over-interpreted; the comparisons that the data support are relative ones between pipelines evaluated on identical input. The wall-thickness study is an inter-method and reproducibility exercise, not a clinical validation. Without expert manual wall-thickness measurements, this work can claim an objective and locally resolved computational framework, but not the replacement of clinical measurement protocols.

Several further factors bound the external validity of the results, and the whole evaluation should be read as preliminary. The general evaluation uses held-out patients from ACDC and M&Ms-2, whereas the pathology-stratified regional analysis is limited to 30 ACDC test patients: 20 normal and 10 with hypertrophic cardiomyopathy. This subset is far too small to support claims of clinical or pathological generalisation, and the hypertrophic subgroup in particular is only large enough for a directional observation. The regional analysis carries a further dependency: each patient contributes seventeen segments, so the segment values are repeated measures within a patient rather than independent samples, and they are reported as a descriptive summary for that reason. The relative thickening figures deserve particular caution, because dividing a systolic by a diastolic thickness amplifies small differences in where the wall is sampled. The regional summaries also depend on the AHA orientation, while all mesh-based measurements depend on the inference grid. A finer grid or an alternative anatomical frame could therefore shift the values. Only the decoder-offset map in Figure 4.3 is calibrated to the segmentation reference; the quantitative tables report geometric estimators before calibration. The reconstruction additionally inherits the quality of the input segmentation, and the training data span more than one collection, so some domain shift between acquisition sources is expected. Finally, the synthetic epicardium used during pre-training is a geometric construction rather than a measured surface, which the fine-tuning stage on real data is designed to correct but cannot fully rule out as a residual prior. A larger cohort, and ideally a dataset with genuine 3D ground truth, would be needed to confirm the results reported here.

The comparison between reconstructions is bounded in the same way. The implicit fit and the SSM fit were chosen because their priors differ from each other and from the learned one, so the spread between the three bounds the influence of the prior; it does not identify which is closest to the anatomy. The implicit fit is moreover of the same family as the surrogate targets used to fine-tune the model, so its agreement with the model is not an independent check. Contour fidelity cannot settle the question either, because it evaluates only the observed rings and not the geometry between them. The slice-decimation analysis of Table 4.3 is bounded more tightly still: it is a single end-diastolic case, so it indicates the direction and rough magnitude of the dependence on slice density but does not estimate it for the cohort. It also measures self-consistency within a method, which rewards a method whose prior ignores the data as much as one that generalises well from it. The comparison does not cover spline lofting, Poisson reconstruction, or other learned models, and the SSM comparison is limited to end-diastole because the public model provides no end-systolic modes. No controlled ablation removes synthetic pre-training, local conditioning, phase conditioning, or positive surface coupling. The experiment therefore characterises the model against two conventional alternatives, but it cannot attribute its behaviour to a specific component or establish that the full architecture is necessary.

CHAPTER 5

Conclusions

This thesis investigated the reconstruction of continuous three-dimensional left ventricular (LV) geometry from sparse short-axis (SAX) contours with an implicit neural representation. It also examined whether the reconstructed surfaces support computational estimates of myocardial wall thickness. The evaluation establishes what the proposed approach achieves and where the present evidence remains limited.

5.1. Answers to the Research Questions

For **RQ1**, the model reconstructs continuous endocardial and epicardial surfaces from the held-out ACDC and M&Ms-2 contours, with contour fidelity near the in-plane image resolution and watertightness after repair. Differences among the three reconstruction methods show that the completed wall geometry remains sensitive to the chosen prior.

For **RQ2**, one phase-conditioned model represents both surfaces at ED and ES while maintaining positive separation between their implicit fields. The lack of matched ES modes for the shape-model comparator limits this finding to phase-conditioned representation rather than phase-specific accuracy.

For **RQ3**, field-based estimates on the proposed and RBF-derived geometries preserve similar between-patient variation but differ in absolute scale, while the cone-ray estimator diverges more strongly. Wall-thickness values therefore depend on both the reconstruction prior and the measurement path.

5.2. Contributions

The first contribution is a common contour-and-surface data pipeline for synthetic LV shapes and clinical ED/ES segmentations. The second is a phase-conditioned implicit model that combines global and local contour features and couples the epicardial field to the endocardial field through a positive offset. The third is an evaluation framework that connects surface and volume metrics with four local thickness estimators, continuous 3D maps, AHA-17 regional summaries, and patient-level NOR–HCM comparisons. Together, these components provide a reproducible way to study geometric completion and derived measurements from sparse SAX observations.

5.3. Limitations

As detailed in Section 4.5, no independent 3D ground truth exists for these scans, so the reconstruction is characterised by contour fidelity and by agreement with alternative reconstructions rather than by accuracy. Mesh repair contributes to the final watertight surfaces, and calibrated decoder-offset figures inherit their absolute scale from the segmentation reference. The cohort used for the pathology analysis is limited to 20 NOR and 10 HCM cases from the ACDC test set, and the group analysis is descriptive.

Finally, no controlled architectural ablation was completed, so the contribution of each model component has not been isolated.

5.4. Future Work

The first priority is evaluation against independent anatomical evidence, such as higher-resolution volumetric imaging or expert 3D surface and wall-thickness annotations, followed by validation on a larger external cohort. The second is a broader matched comparison, extending the present implicit and shape-model comparators with spline-based lofting and learned alternatives, using the same patients, physical spacing, repair procedure, and metrics, together with point-wise error maps that localise failures at the apex, base, and highly curved wall regions. The third is a targeted experimental study of synthetic pre-training, local conditioning, positive surface coupling, and wall-offset supervision. Extending phase conditioning beyond a binary ED/ES label would then allow temporal consistency across the full cardiac cycle to be investigated.

Overall, the thesis establishes a reproducible framework for reconstructing paired LV surfaces and deriving locally resolved thickness summaries from sparse SAX contours.

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APPENDIX A

Supplementary Methodological Details

This appendix summarises the training streams, data partitions, clinical RBF mesh-preparation operations, and model configuration described in Chapter 3.

Table A.1. Training data streams and their roles.

Stream	Phase	Source	Training role
Synthetic SSM	ED	UK Digital Heart Project LV SSM	Pre-training to establish a geometric shape prior
Real MRI	ED	ACDC and M&Ms-2	Fine-tuning on patient-specific diastolic anatomy
Real MRI	ES	ACDC and M&Ms-2	Fine-tuning on systolic anatomy with phase conditioning

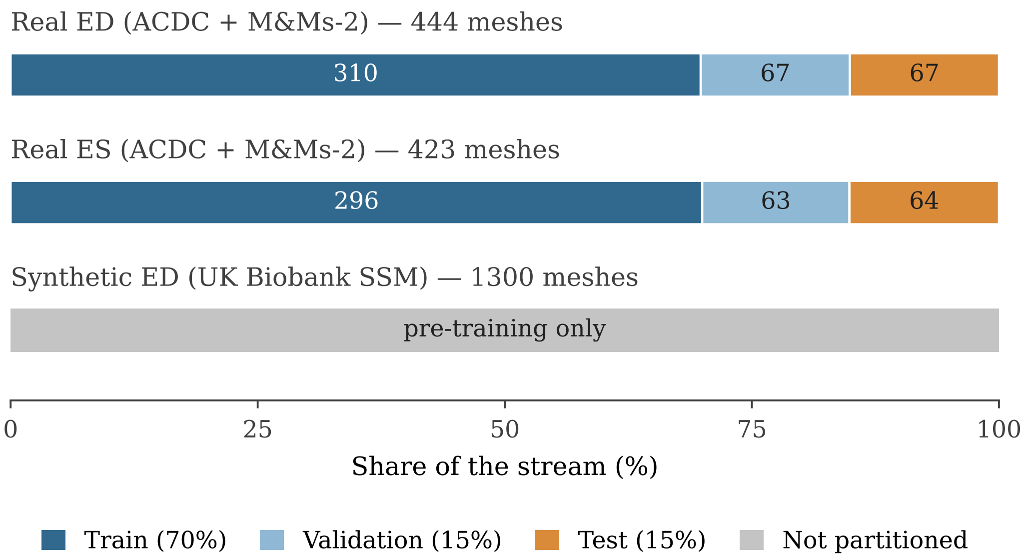

Figure A.1. Valid clinical samples divided into the 70 %/15 %/15 % partitions. The 1300 accepted synthetic meshes are reserved for pre-training.

Table A.2. Operations used to construct the RBF-derived clinical supervision meshes.

Operation	Purpose
RAS+ canonicalisation	Standardises patient-coordinate orientation while preserving physical spacing
Label harmonisation	Maps ACDC and M&Ms-2 labels to common endocardial and epicardial definitions
Largest-ring extraction and resampling	Retains 60 ordered points per boundary on each acquired SAX slice
In-plane normal estimation	Defines an outward direction at every contour point
Off-surface constraints at ± 2.5 mm	Establishes the sign and local scale of each implicit field
Thin-plate-spline RBF fit	Interpolates a continuous field with degree 1 and regularisation parameter 0.5
Evaluation on a 1.0 mm grid	Discretises the continuous field for surface extraction
Signed-distance regularisation ($\sigma = 1.2$ mm)	Reduces voxel-scale corrugation before surface extraction
Connected-region filtering and marching cubes	Removes invalid regions and extracts the zero level set
Topology validation and conditional correction	Preserves valid RBF surfaces and corrects only failed shells
Geometric plausibility checks	Rejects failed fits and invalid endocardial–epicardial pairs

Table A.3. Main architecture settings of the model.

Component	Configuration
Input features	3D coordinates, tissue label, and cardiac phase
Global encoder type	Shared network with endocardial and epicardial max-pooling
Local conditioning	Coarse 3D contour feature volume with trilinear interpolation
Latent dimension	256
Local volume size	16^3 with 32 channels
Positional encoding	Fourier features with $L = 3$ frequency bands
Decoder	8-layer network, width 512, skip connection at layer 4
Activation	Softplus (smooth)
Local injections	Local feature paths into the decoder trunk and offset head
Output heads	Endocardial signed distance and positive decoder wall-offset field
Offset mapping	Sigmoid component with $\tau_{\text{min}} = 0.05$ and $\delta_{\text{cap}} = 0.45$, plus learned soft-hinge headroom
Inference grid	96^3 query grid for marching cubes